

* For modulus e.g. $|-3|$ we write magnitude only not sign i.e. 3 in this case.

Important Formulas

- 1). $(a + b)^2 = a^2 + 2ab + b^2$
- 2). $(a - b)^2 = a^2 - 2ab + b^2$
- 3). $(a^2 - b^2) = (a + b)(a - b)$
- 4). $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- 5). $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- 6). $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
- 7). $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
- 8). $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
 $= \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$
 Here, if $(a + b + c) = 0$,
 then $a^3 + b^3 + c^3 - 3abc = 0$
 $\Rightarrow a^3 + b^3 + c^3 = 3abc$
- 9). $(a + b + c)^2 = (a^2 + b^2 + c^2) + 2(ab + bc + ca)$

Variety Questions

Q.1. What is the value of

$$\sqrt{20 + \sqrt{20 + \sqrt{20 + \dots}}}$$

SSC CGL 17/09/2025 (Shift 1)

- (a) 4 (b) 7 (c) 5 (d) 6

Q.2. Choose the incorrect relation from the following:

I. $\sqrt{7} + \sqrt{3} = \sqrt{6} + \sqrt{3}$

II. $\sqrt{7} + \sqrt{3} < \sqrt{6} + \sqrt{3}$

III. $\sqrt{7} + \sqrt{3} > \sqrt{6} + \sqrt{3}$

Graduate Level 29/08/2025 (Shift - 1)

- (a) II (b) I and II (c) I (d) II and III

Q.3. The value of: $(2 + \sqrt{3}) + (\frac{1}{2 - \sqrt{3}}) -$

$(4 + 2\sqrt{3})$ is:

Graduate Level 29/07/2025 (Shift - 3)

- (a) 0 (b) 2 (c) 6 (d) $\sqrt{2}$

Q.4. Find the value of: $(\sqrt{576} - \sqrt{144}) \times (\sqrt{64} + \sqrt{225})$

Higher Secondary 29/07/2025 (Shift - 2)

- (a) 276 (b) 270 (c) 285 (d) 250

Q.5. Which fraction is equivalent to 0.365?

Matriculation Level 29/07/2025 (Shift - 1)

- (a) $\frac{365}{10}$ (b) $\frac{37}{50}$ (c) $\frac{73}{200}$ (d) $\frac{75}{200}$

Q.6. Simplify $\frac{(25)^4 + 1250 + 1 - (25)^2}{(25)^2 + 25 + 1}$.

- (a) 602 (b) 601 (c) 603 (d) 604

Q.7. Simplify

$$\frac{25a^2 - 10ab - 48b^2}{5a + 6b} \times (5a + 8b)$$

- (a) $25a^2 - 64b^2$ (b) $(5a - 8b)^2$
 (c) $(5a + 8b)^2$ (d) $25a^2 - 36b^2$

Q.8. The simplified form of the expression

$$\frac{(x^2 - 9) \times (9x^2 - 1)}{(x - \frac{1}{3}) \times (1 - \frac{x}{3})} \times \frac{9}{(x + \frac{1}{3})(1 + \frac{x}{3})}$$

is _____.

- (a) -729 (b) -81 (c) 729 (d) 81

Q.9. If $A = \frac{58^2 - 25^2}{46^2 - 37^2}$, $B = \frac{26^2 - 15^2}{56^2 - 15^2}$,

then the value of $\frac{1}{B} - \frac{20}{A}$ is :

- (a) 2 (b) 1 (c) -1 (d) 0

Q.10. On comparing the following two numeric expressions, we find that _____.

$$[(2\frac{7}{9})^2]^{\frac{3}{5}} \text{ and } [(1\frac{2}{3})^5]^{\frac{3}{5}}$$

- (a) both the expressions are equal
 (b) the first expression is smaller than the second
 (c) the first expression is larger than the second
 (d) the given two expressions cannot be compared

Q.11. On simplifying the following expression, we get _____.

$$\frac{24\sqrt{45} - 16\sqrt{20} + \sqrt{245} - 47\sqrt{5}}{3\frac{1}{5} + 1\frac{2}{5} - 2\frac{1}{6} - 2\frac{1}{5} - 5\frac{2}{5} + 3\frac{1}{6}}$$

- (a) 0 (b) 3 (c) 1 (d) 2

Q.12. Simplify :

$$\frac{(3.321)^3 + (2.681)^3 + (1.245)^3 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

- (a) 6.125 (b) 8.645 (c) 7.247 (d) 10.245

Q.13. Simplify the following expression.

$$0.01 + 0.68 \times 0.68 + 0.22 \times 0.22 + 0.044 + 0.136 + 0.44 \times 0.68$$

- (a) 0.69 (b) 1 (c) 0.8 (d) 0.9

Q.14. Simplify the given expression .

$$\frac{0.09 \times 0.09 + 0.04 \times 0.16 + 0.16 \times 0.16 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.3 \times 0.3 + 0.2 \times 0.2 + 0.4 \times 0.4}$$

- (a) 0.38 (b) 0.24 (c) 0.32 (d) 0.29

Q.15. What is the value of the given Expression ?

$$\frac{4^{a+4} - 5 \times 4^{a+2}}{15 \times 4^a - 2^2 \times 4^a}$$

- (a) 16 (b) 64 (c) 20 (d) 24

Q.16. What is the value of $(2^2 + 1)(2^4 + 1)$

$$(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1) ?$$

- (a) $\frac{2^{256} - 1}{2}$ (b) $\frac{2^{256} - 1}{3}$
 (c) $2^{256} - 1$ (d) $\frac{2^{256} - 1}{4}$

Q.17. Which of the following statement is true ?

I. $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{110} < \frac{5}{6}$

II. $\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{143} > \frac{7}{13}$

- (a) Only I (b) Both I and II
 (c) Only II (d) Neither I nor II

Q.18. Find the value of the following expression.

$$[\frac{5}{8} - \{\frac{3}{8} - (\frac{5}{8} - \frac{3}{8})\}] \text{ of } 8.8 - 1.2$$

$$4\frac{1}{6} \div 2.5 \times 2 \div \frac{1}{6} \text{ of } 60 + (\frac{3}{4} - \frac{3}{8})$$

- (a) $5\frac{22}{43}$ (b) $3\frac{23}{67}$ (c) $4\frac{44}{85}$ (d) $4\frac{4}{5}$

Q.19. Solve the following.

$$24 \div \frac{3}{8} \text{ of } (8 + 2 \times \overline{7-3}) + [\frac{2}{11} \div \frac{4}{55} - \{\frac{5}{8} + \frac{6}{16}\}]$$

- (a) $\frac{23}{27}$ (b) $\frac{9}{2}$ (c) $\frac{11}{18}$ (d) $\frac{15}{19}$

Q.20. Solve

$$\frac{(6.9)^3 + (2.7)^3 + (5.8)^3 - 3 \times 6.9 \times 2.7 \times 5.8}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)}$$

- (a) 15.1 (b) 15.3 (c) 15.2 (d) 15.4

Q.21. Simplify the following expression.

$$9\frac{1}{5} \div \{(16\frac{1}{5} \div 12\frac{2}{3} - 9\frac{2}{3}) +$$

$$17\frac{1}{2} \text{ of } 3\frac{1}{21}\}$$

- (a) $\frac{139}{881}$ (b) $\frac{137}{881}$ (c) $\frac{140}{881}$ (d) $\frac{138}{881}$

Q.22. The value of

$$\frac{1}{4} + \frac{[(20.35)^2 - (8.35)^2] \times 0.0175}{(1.05)^2 + (1.05)(27.65)}$$
 is:

- (a) $\frac{3}{20}$ (b) $\frac{3}{10}$ (c) $\frac{7}{20}$ (d) $\frac{9}{20}$

Q.23. The simplified value of

$$\frac{119 - [48 \div 6 - 7 \{5 \times 12 \div 3 - (15 - \overline{3-8})\}]}{24 \div 3 - 9 \text{ of } 3 + (52 - 8 - 4) \div 6}$$

is:

- (a) $-\frac{111}{11}$ (b) $-\frac{127}{27}$ (c) $\frac{127}{43}$ (d) $\frac{111}{11}$

Q.24. If $\frac{1}{4 - \sqrt{8}} + \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} - \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}}$

$= a + b\sqrt{2}$, then what is the value of $(3a + 4b)$?

- (a) $99\frac{1}{2}$ (b) 97 (c) $98\frac{1}{2}$ (d) 98

Q.25. If the following interchanges are made in signs and numbers, which equation would be correct ?

Interchanges: Signs: $\div$ and $-$ Numbers: 15 and 5

(a) $4 \times 30 - 15 \div 5 + 12 = 21$

(b) $12 \times 30 - 15 \div 5 + 4 = 37$

(c) $4 \times 30 - 15 \div 5 + 12 = 27$

(d) $4 \times 30 - 5 \div 15 + 14 = 21$



Q.26. If $\frac{4}{1 + \sqrt{2} + \sqrt{3}} = a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$, where a, b, c, d are natural numbers, then the value of a + b + c + d is:
(a) 0 (b) 2 (c) 4 (d) 1

Q.27.
If $\left[\left(\frac{2}{3}\right)^3(2x+3)\right]^{\frac{-3}{4}} = \left[\left(\frac{2}{3}\right)^{\frac{2}{3}}(3x+7)\right]^{\frac{-6}{5}}$
then the value of $\sqrt{2 - 42x}$ is:
(a) 5 (b) 6 (c) 3 (d) 4

Q.28. If '+' means '-', '-' means '+', 'x' means '÷' and '÷' means 'x' then the value of $\frac{42 - 12 \times 3 + 8 \div 2 + 15}{8 \times 2 - 4 + 9 \div 3}$ is:
(a) $-\frac{5}{3}$ (b) $-\frac{15}{19}$ (c) $\frac{15}{19}$ (d) $\frac{5}{3}$

Q.29. The value of $\frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$ is:
(a) 24.42 (b) 24.24 (c) 25.42 (d) 25.48

Q.30. The value of $\frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$ is:
(a) -1 (b) 1 (c) 3 (d) -3

Q.31. Find the value $\left\{ \frac{(0.7)^2 \div 0.14 + (0.6)^2 \div 0.18 + (0.5)^2 \div 0.05}{4(2.5 \text{ of } 4 - 13 \times 0.25 \times 3)} \right\}$
(a) $\frac{25}{2}$ (b) $\frac{19}{2}$ (c) $\frac{23}{2}$ (d) $\frac{21}{2}$

Q.32. If $x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} + \dots + \frac{1}{23.24}$, $y = \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} + \dots + \frac{1}{71.72}$ then $\frac{x}{y}$ is equal to:
(a) $\frac{1}{3}$ (b) $\frac{1}{24}$ (c) $\frac{1}{72}$ (d) 3

Q.33. The simplified value of $\frac{0.01404}{24^2 + 6^2 - 144}$ is:
(a) 3×10^{-5} (b) 6×10^{-5}
(c) 2.4×10^{-4} (d) 3×10^{-4}

Q.34. The simplified value of $\frac{1.0025 + 6.25 \times 10^{-6}}{0.0025 + 0.95}$ is:
(a) 1.0025 (b) 1.0525
(c) 1.0005 (d) 1.0505

Examination wise Questions

SSC CGL 2025 Tier - I

Q.35. Simplify: $(3\frac{1}{2} + 4.5) - 1.9$
SSC CGL 12/09/2025 (Shift 1)

(a) 5.2 (b) 4.3 (c) 6.1 (d) 3.2

Q.36. Evaluate:
 $27\frac{1}{4} - [\frac{13}{6} \div \{\frac{1}{3} - (\frac{1}{2} \times (\frac{3}{4} - \frac{1}{4}))\}]$
SSC CGL 12/09/2025 (Shift 1)
(a) $-2\frac{1}{4}$ (b) $2\frac{1}{4}$ (c) $1\frac{1}{4}$ (d) $1\frac{3}{4}$

Q.37. $32^3 + 15^3 - 35^3 + 250$ is equal to:
SSC CGL 12/09/2025 (Shift 1)
(a) -6582 (b) -6482 (c) -6682 (d) -5482

Q.38. Evaluate the continued fraction:
 $x = 4 + \frac{1}{5 + \frac{1}{6 + \frac{1}{2}}}$
SSC CGL 12/09/2025 (Shift 2)
(a) $\frac{281}{67}$ (b) $\frac{283}{67}$ (c) $\frac{286}{67}$ (d) $\frac{287}{67}$

Q.39. Simplify:
 $(\frac{1}{5 + \sqrt{3}}) + (\frac{1}{5 - \sqrt{3}}) - (\frac{10}{87 - 65})$
SSC CGL 12/09/2025 (Shift 3)
(a) 2 (b) 1 (c) 0 (d) $\sqrt{3}$

Q.40. Evaluate:
 $12(\frac{1}{2})\% \text{ of } 480 \text{ km} + 66(\frac{2}{3})\% \text{ of } 270 \text{ km}$
SSC CGL 12/09/2025 (Shift 3)
(a) 240 km (b) 230 km
(c) 320 km (d) 450 km

Q.41. Evaluate:
 $6 - [\frac{3}{4} \div \{\frac{1}{5} + (\frac{1}{2} \times (1 - \frac{1}{3}))\}]$
SSC CGL 13/09/2025 (Shift 1)
(a) $\frac{14}{3}$ (b) $\frac{157}{32}$ (c) $\frac{147}{32}$ (d) $\frac{17}{3}$

Q.42. What is the value of $7 + 77 + 777 + 7777 + 77777$?
SSC CGL 13/09/2025 (Shift 2)
(a) 86415 (b) 86385 (c) 86420 (d) 86500

Q.43. Simplify: $(1\frac{4}{5} + 3.2) - 1\frac{1}{4}$
SSC CGL 13/09/2025 (Shift 2)
(a) 3.75 (b) 2.1 (c) 2.35 (d) 2.5

Q.44. Find the product: $1.2 \times 2\frac{2}{5} \times \frac{3}{6}$
SSC CGL 14/09/2025 (Shift 1)
(a) 2.4 (b) 1.5 (c) 1.44 (d) 2.1

Q.45. $0.2 + 0.02 + 0.002 = ?$
SSC CGL 14/09/2025 (Shift 1)
(a) 0.223 (b) 0.111 (c) 0.221 (d) 0.222

Q.46. What is the result of $3\frac{1}{4} \div 0.5$?
SSC CGL 14/09/2025 (Shift 2)
(a) 3.25 (b) 6.5 (c) 2.75 (d) 6.25

Q.47. Evaluate the expression:
 $(\frac{1}{9} \div \frac{1}{3}) \div (\frac{1}{15} + 1 - \frac{9}{10})$

SSC CGL 14/09/2025 (Shift 3)
(a) 5 (b) 6 (c) 3 (d) 2

Q.48. Simplify: $\sqrt{98} + \sqrt{32} - \sqrt{50}$
SSC CGL 14/09/2025 (Shift 3)
(a) $7\sqrt{2}$ (b) $6\sqrt{2}$ (c) $5\sqrt{2}$ (d) $8\sqrt{2}$

Q.49. Simplify: $(\frac{8}{5} \div \frac{14}{5}) \times (\frac{10}{7} \div \frac{4}{3})$
SSC CGL 15/09/2025 (Shift 1)
(a) $\frac{120}{196}$ (b) $\frac{180}{196}$ (c) $\frac{196}{180}$ (d) $\frac{150}{360}$

Q.50. Simplify: $(3.5 + \frac{3}{8}) - (\frac{9}{5} \times 1.25)$
SSC CGL 15/09/2025 (Shift 1)
(a) 3.4418 (b) 1.625
(c) 4.5526 (d) 7.5526

Q.51. Simplify:
 $((1.5 + \frac{1}{6}) \div (2.25 - \frac{1}{3})) + 0.8$
SSC CGL 15/09/2025 (Shift 1)
(a) 1.88 (b) 2.88 (c) 4.88 (d) 1.66

Q.52. What is the value of $(\frac{14}{16} \times 4.8) \div 0.12$?
SSC CGL 15/09/2025 (Shift 2)
(a) 35 (b) 25 (c) 32 (d) 28

Q.53. Evaluate: $40\% \text{ of } 175 - 37(\frac{1}{2})\% \text{ of } 160 + 10\% \text{ of } 250$
SSC CGL 15/09/2025 (Shift 2)
(a) 25 (b) 50 (c) 35 (d) 30

Q.54. Simplify: $\sqrt{196} - \sqrt{64} + \sqrt{100}$
SSC CGL 15/09/2025 (Shift 2)
(a) 20 (b) 16 (c) 15 (d) 18

Q.55. Simplify: $(3.2 - 1\frac{1}{5}) + (2\frac{1}{4} \div 0.5)$
SSC CGL 15/09/2025 (Shift 3)
(a) 9.2 (b) 6.5 (c) 8.7 (d) 9.5

Q.56. Which of the following represents the correct simplified value of the continued fraction: $x = 1 + \frac{1}{3 + \frac{1}{1 + \frac{1}{2}}}$

SSC CGL 15/09/2025 (Shift 3)
(a) $\frac{14}{11}$ (b) $\frac{11}{12}$ (c) $\frac{11}{5}$ (d) $\frac{9}{7}$

Q.57. $2.5\% \text{ of } 100 = ?$
SSC CGL 15/09/2025 (Shift 3)
(a) 6 (b) 2 (c) 3 (d) 2.5

Q.58. Simplify:
 $[(\frac{4}{3}) + (\frac{4}{5} \div \frac{2}{7})] \div [(\frac{3}{2}) - (\frac{7}{12} \times \frac{3}{4})]$
SSC CGL 16/09/2025 (Shift 1)
(a) $\frac{992}{255}$ (b) $\frac{892}{255}$ (c) $\frac{982}{255}$ (d) $\frac{929}{255}$

Q.59. Simplify:



$$(2\frac{1}{2} + 2.6) \div 0.75 + (1.25 \times \frac{3}{5})$$

SSC CGL 16/09/2025 (Shift 1)

(a) $\frac{137}{20}$ (b) $\frac{161}{20}$ (c) $\frac{151}{20}$ (d) $\frac{148}{20}$

Q.60. Simplify: $(\frac{5}{6} + \frac{3}{4}) \div (\frac{7}{8} - \frac{2}{3})$

SSC CGL 16/09/2025 (Shift 2)

(a) $\frac{38}{5}$ (b) $\frac{48}{5}$ (c) $\frac{58}{5}$ (d) $\frac{68}{5}$

Q.61. Simplify: $\sqrt{45} + \sqrt{20} - \sqrt{80}$.

SSC CGL 16/09/2025 (Shift 2)

(a) $4\sqrt{5}$ (b) $6\sqrt{5}$ (c) $\sqrt{5}$ (d) $7\sqrt{5}$

Q.62. Evaluate:

$$(0.04^3 + 0.01^3) \div (0.16^3 + 0.04^3)$$

SSC CGL 16/09/2025 (Shift 2)

(a) 0.008 (b) 0.015 (c) 0.016 (d) 0.009

Q.63. Convert 4.625 into a fraction in simplest form.

SSC CGL 16/09/2025 (Shift 3)

(a) $\frac{37}{8}$ (b) $\frac{27}{4}$ (c) $\frac{38}{7}$ (d) $\frac{25}{6}$

Q.64. Find the difference:

$$7.4 - (2\frac{7}{8} + 2.75)$$

SSC CGL 16/09/2025 (Shift 3)

(a) 7.822 (b) 1.775 (c) 1.522 (d) 2.025

Q.65. Add $(\frac{1}{16} + \frac{1}{32})$ and express the result as a decimal.

SSC CGL 16/09/2025 (Shift 3)

(a) 0.09275 (b) 0.09375
(c) 0.09175 (d) 0.09475

Q.66. Simplify: $\sqrt{(14 + 6\sqrt{5})}$

SSC CGL 17/09/2025 (Shift 1)

(a) $2 + \sqrt{5}$ (b) $3 + \sqrt{5}$
(c) $\sqrt{3} + 2\sqrt{5}$ (d) $9 + \sqrt{5}$

Q.67. $42^3 + 45^3 - 63^3 + 149$ is equal to:

SSC CGL 17/09/2025 (Shift 1)

(a) -84685 (b) 0 (c) -78659 (d) 1

Q.68. Simplify:

$$(2.8 + 5.125) \times 0.45 - (0.90 \div 0.3)$$

SSC CGL 17/09/2025 (Shift 2)

(a) 0.566 (b) 0.11875
(c) 2.8756 (d) -2.8756

Q.69. What is the value of

$$(0.03^3 + 0.002^3) \div (0.3^3 + 0.02^3) ?$$

SSC CGL 17/09/2025 (Shift 2)

(a) 0.002 (b) 0.0045
(c) 0.0102 (d) 0.001

Q.70. Evaluate $\sqrt{30 + \sqrt{30 + \sqrt{30 + \dots}}}$

SSC CGL 17/09/2025 (Shift 3)

(a) 3 (b) 5 (c) 4 (d) 6

Q.71. $18^3 + 20^3 - 29^3 + 116$ is equal to:

SSC CGL 17/09/2025 (Shift 3)

(a) -10441 (b) -16789 (c) 0 (d) 1

Q.72. What will come in place of ? to satisfy the equation: $(\sqrt{5} + 3)^2 = ? + 6\sqrt{5}$

SSC CGL 18/09/2025 (Shift 1)

(a) 17 (b) 16 (c) 15 (d) 14

Q.73. Solve: $(\sqrt{7,396} - \sqrt{1,156}) \times (\frac{1}{2})$

SSC CGL 18/09/2025 (Shift 1)

(a) 22 (b) 23 (c) 26 (d) 27

Q.74. What is the value of

$$(0.11^3 + 0.08^3) \div (0.22^3 + 0.16^3) ?$$

SSC CGL 18/09/2025 (Shift 1)

(a) 0.125 (b) 0.5 (c) 0.25 (d) 1

Q.75. Simplify: $\{(0.6)^3 + (0.06)^3 + (0.006)^3\} \div \{(0.2)^3 + (0.02)^3 + (0.002)^3\}$

SSC CGL 18/09/2025 (Shift 1)

(a) 5 (b) 25 (c) 27 (d) 17

Q.76. Find the value of

$$\sqrt{(711 + \sqrt{(303 + \sqrt{441})})}$$

SSC CGL 18/09/2025 (Shift 2)

(a) 44 (b) 37 (c) 27 (d) 33

Q.77. Evaluate:

$$(0.09^3 + 0.03^3) \div (0.3^3 + 0.1^3)$$

SSC CGL 18/09/2025 (Shift 2)

(a) 0.028 (b) 0.056 (c) 0.003 (d) 0.027

Q.78. Solve: $(\sqrt{6400} - \sqrt{2401}) \times (\frac{1}{2})$

SSC CGL 18/09/2025 (Shift 3)

(a) 17.75 (b) 18.5 (c) 15.5 (d) 10.75

Q.79. Solve: $(\sqrt{5776} - \sqrt{4761}) \times (\frac{1}{3})$

SSC CGL 18/09/2025 (Shift 3)

(a) 3.5 (b) 4.3 (c) 3.3 (d) 2.3

Q.80. Simplify:

$$(0.16^3 + 0.04^3) \div (0.4^3 + 0.1^3)$$

SSC CGL 18/09/2025 (Shift 3)

(a) 0.64 (b) 0.064 (c) 0.6 (d) 0.006

Q.81. Find the value of $(0.02^3 + 0.002^3) \div (0.04^3 + 0.004^3)$

SSC CGL 18/09/2025 (Shift 3)

(a) $\frac{1}{8}$ (b) $\frac{1}{16}$ (c) $\frac{1}{64}$ (d) $\frac{1}{128}$

Q.82. Simplify the expression:

$$\sqrt{(10 + 4\sqrt{6})}$$

SSC CGL 19/09/2025 (Shift 1)

(a) $\sqrt{6} + 2$ (b) $3 + \sqrt{5}$
(c) $2 + \sqrt{6}$ (d) $3 + \sqrt{4}$

Q.83. $17^3 + 20^3 - 19^3 + 346$ is equal to:

SSC CGL 19/09/2025 (Shift 1)

(a) 0 (b) 6400 (c) -4860 (d) 5200

Q.84. What is the result of $(0.4^3 + 0.3^3 - 0.7^3) \div (3 \times 0.4 \times 0.3 \times 0.7) ?$

SSC CGL 19/09/2025 (Shift 1)

(a) -1 (b) $\frac{7}{9}$ (c) $\frac{8}{9}$ (d) $\frac{6}{9}$

Q.85. Evaluate: $(\sqrt{13} + \sqrt{5})^2 - (\sqrt{13} - \sqrt{5})^2$

SSC CGL 19/09/2025 (Shift 2)

(a) $4\sqrt{65}$ (b) $2\sqrt{65}$ (c) $6\sqrt{65}$ (d) $8\sqrt{65}$

Q.86. If $p = 0.7$ and $q = 0.05$, find the

value of $\frac{(3p)^3 + (3q)^3}{p^3 + q^3}$.

SSC CGL 19/09/2025 (Shift 2)

(a) 64 (b) 27 (c) 59 (d) 65

Q.87. Simplify: $\frac{1}{\sqrt{5} - \sqrt{2}} + \frac{1}{\sqrt{5} + \sqrt{2}}$

SSC CGL 19/09/2025 (Shift 3)

(a) $\sqrt{5}$ (b) $\frac{5\sqrt{2}}{3}$ (c) $\frac{2\sqrt{5}}{7}$ (d) $\frac{2\sqrt{5}}{3}$

Q.88. What is the value of $(0.3^3 + 0.03^3) \div (0.6^3 + 0.06^3) ?$

SSC CGL 19/09/2025 (Shift 3)

(a) 0.065 (b) 0.005 (c) 0.225 (d) 0.125

Q.89. Multiply 14.345 by 0.037 and round your answer to 4 decimal places.

SSC CGL 20/09/2025 (Shift 1)

(a) 0.5307 (b) 0.5370
(c) 0.5037 (d) 0.5703

Q.90. What is the value of $\sqrt{9409} - \sqrt{1849}$

SSC CGL 20/09/2025 (Shift 2)

(a) 54 (b) 68 (c) 78 (d) 88

Q.91. Simplify: $\sqrt{20} + \sqrt{80}$

SSC CGL 20/09/2025 (Shift 2)

(a) $6\sqrt{5}$ (b) $7\sqrt{5}$ (c) $8\sqrt{5}$ (d) $9\sqrt{5}$

Q.92. Add: $8.65 + 9.28 + 0.067$

SSC CGL 20/09/2025 (Shift 3)

(a) 11.257 (b) 8.867 (c) 7.768 (d) 17.997

Q.93. The value of

$$\sqrt{13 - 2\sqrt{42}} - \frac{1}{\sqrt{13 - 2\sqrt{42}}}$$
 is:

SSC CGL 21/09/2025 (Shift 2)

(a) $1 + \sqrt{5}$ (b) $2\sqrt{6}$ (c) $-2\sqrt{6}$ (d) $1 - \sqrt{5}$

Q.94. What is the simplified value of:

$$\frac{0.02505}{23^2 + 6^2 - 64}$$

SSC CGL 21/09/2025 (Shift 2)

(a) 5×10^{-5} (b) 3×10^{-5}
(c) 2.4×10^{-4} (d) 3×10^{-4}

Q.95. What will come in place of ? to satisfy the equation: $(\sqrt{8} - 1)^2 = ? - 2\sqrt{8}$

SSC CGL 21/09/2025 (Shift 3)

(a) 9 (b) 3 (c) 5 (d) 4

Q.96. Simplify: $\frac{\sqrt{7+4\sqrt{3}}}{\sqrt{7-4\sqrt{3}}}$

SSC CGL 21/09/2025 (Shift 3)

(a) $7 - 4\sqrt{3}$ (b) $7 + 4\sqrt{3}$



(c) $3 + 2\sqrt{3}$ (d) $3 - 2\sqrt{3}$

Q.97. Which of the following numbers has a terminating decimal representation?

SSC CGL 22/09/2025 (Shift 1)

(a) $\frac{6}{16}$ (b) $\frac{2}{12}$ (c) $\frac{4}{14}$ (d) $\frac{8}{18}$

Q.98. What will come in place of ? to satisfy the equation:

$$(\sqrt{7} - 2)^2 = ? - 4\sqrt{7}$$

SSC CGL 22/09/2025 (Shift 2)

(a) 12 (b) 11 (c) 10 (d) 13

Q.99. Simplify: $(\sqrt{5} + \sqrt{3})^2$

SSC CGL 22/09/2025 (Shift 2)

(a) $8 + 2\sqrt{15}$ (b) $8 - 2\sqrt{15}$

(c) $3 + 2\sqrt{15}$ (d) $3 - 2\sqrt{15}$

Q.100. Simplify:

$$\frac{\sqrt{16+2\sqrt{55}} + \sqrt{16-2\sqrt{55}}}{\sqrt{4}}$$

SSC CGL 22/09/2025 (Shift 3)

(a) $2\sqrt{3}$ (b) $\sqrt{11}$ (c) $3\sqrt{2}$ (d) $\sqrt{5}$

Q.101. Solve:

$$\sqrt{4 \times 225 \div 15 + 25 - 4}$$

SSC CGL 23/09/2025 (Shift 2)

(a) $\sqrt{81}$ (b) $\sqrt{21}$ (c) $\sqrt{71}$ (d) $\sqrt{41}$

Q.102. Simplify : $\sqrt{(11 + 4\sqrt{6})}$

SSC CGL 24/09/2025 (Shift 1)

(a) $2\sqrt{3} + 3$ (b) $2\sqrt{2} + \sqrt{3}$

(c) $\sqrt{11} + \sqrt{5}$ (d) $2\sqrt{3} + \sqrt{2}$

Q.103. What will take the place of "?" in

the given equation: $\frac{?}{\sqrt{64}} = \frac{\sqrt{121}}{10 + 1}$

SSC CGL 24/09/2025 (Shift 3)

(a) 7 (b) 8 (c) 9 (d) 4

Q.104. Solve: $(\sqrt{6241} - \sqrt{1260.25}) \times (\frac{1}{6})$

SSC CGL 25/09/2025 (Shift 2)

(a) 9.875 (b) 8.875 (c) 7.25 (d) 6.685

Q.105. Simplify: $2\sqrt{5}(\sqrt{45} - \sqrt{20})$

SSC CGL 25/09/2025 (Shift 2)

(a) 6 (b) 4 (c) 5 (d) 10

Q.106. Rationalize: $\frac{\sqrt{6} + \sqrt{7}}{\sqrt{6} - \sqrt{7}}$

SSC CGL 26/09/2025 (Shift 1)

(a) $13 + 2\sqrt{42}$ (b) $-13 - 2\sqrt{42}$

(c) $13 - 2\sqrt{42}$ (d) $-13 + 2\sqrt{42}$

Q.107. Solve :

$$\sqrt{8 \times 447 \div 24 + 75 - 28}$$

SSC CGL 26/09/2025 (Shift 2)

(a) 15 (b) 14 (c) 16 (d) 17

Q.108. Solve: $(12.12)^2 - 396$

SSC CGL 26/09/2025 (Shift 3)

(a) -249.1056 (b) 249.10566

(c) -251.1056 (d) 251.1056

Q.109. Evaluate:

$$9\frac{1}{5} - [\frac{5}{6} \div \{\frac{5}{6} - (\frac{1}{2} \times (\frac{7}{4} - \frac{1}{4}))\}]$$

SSC CGL 14/10/2025 (Shift 1)

(a) $-\frac{4}{5}$ (b) $3\frac{1}{4}$ (c) $-2\frac{3}{4}$ (d) $7\frac{3}{4}$

Q.110. Evaluate the continued fraction:

$$x = 5 + \frac{1}{5 + \frac{1}{6 + \frac{1}{2}}}$$

SSC CGL 14/10/2025 (Shift 1)

(a) $\frac{348}{67}$ (b) $\frac{358}{67}$ (c) $\frac{348}{77}$ (d) $\frac{348}{69}$

Q.111. Evaluate the continued fraction:

$$x = 3 + \frac{1}{8 + \frac{1}{6 + \frac{1}{2}}}$$

SSC CGL 14/10/2025 (Shift 3)

(a) $\frac{331}{106}$ (b) $\frac{131}{106}$ (c) $\frac{321}{106}$ (d) $\frac{231}{106}$

Q.112. Which of the following equals 20 ?

SSC CGL 14/10/2025 (Shift 3)

(a) $(\sqrt{11} + \sqrt{5})^2 - 2\sqrt{55}$

(b) $(\sqrt{5} + \sqrt{5})^2$

(c) $(\sqrt{2} + \sqrt{3})^2 + 2\sqrt{6}$

(d) $(\sqrt{11} - \sqrt{5})^2 + 2\sqrt{55}$

SSC Selection Post (XIII)

Q.113. What is the value of $\frac{(9\sqrt{3})}{(3\sqrt{3})}$?

Matriculation Level 26/07/2025 (Shift - 1)

(a) $\sqrt{3}$ (b) 3 (c) 9 (d) $3\sqrt{3}$

Q.114. Identify the correct decimal

equivalent of $\frac{253}{16}$.

Matriculation Level 26/07/2025 (Shift - 1)

(a) 15.8125 (b) 16.125

(c) 14.9375 (d) 15.9375

Q.115. Find the square root of 0.0049.

Matriculation Level 31/07/2025 (Shift - 1)

(a) 0.007 (b) 0.07 (c) 0.7 (d) 0.0007

Q.116. What is the reciprocal of $\frac{5}{7}$?

Matriculation Level 31/07/2025 (Shift - 1)

(a) $\frac{1}{7}$ (b) $\frac{7}{5}$ (c) $\frac{1}{5}$ (d) $\frac{7}{1}$

Q.117. What is the decimal form of $\frac{9}{20}$?

Matriculation Level 01/08/2025 (Shift - 1)

(a) 0.45 (b) 0.425 (c) 0.5 (d) 0.48

Q.118. The square of 34×34 :

Matriculation Level 01/08/2025 (Shift - 1)

(a) 1024 (b) 1156 (c) 1089 (d) 1296

Q.119. Find the square of 23.

Matriculation Level 01/08/2025 (Shift - 2)

(a) 529 (b) 512 (c) 484 (d) 576

Q.120. Which of the following is the CORRECT comparison ?

Matriculation Level 01/08/2025 (Shift - 2)

(a) $\frac{2}{5} > 0.45$ (b) $0.6 < \frac{3}{4}$

(c) $\frac{7}{10} < 0.65$ (d) $0.3 > \frac{1}{2}$

Q.121. Show that 0.4757575... can be expressed in the form p/q, where p and q are integers and q ≠ 0, What is the value of p/q?

Matriculation Level 01/08/2025 (Shift - 3)

(a) $\frac{157}{330}$ (b) $\frac{151}{330}$ (c) $\frac{129}{310}$ (d) $\frac{164}{330}$

Q.122. Which of the following fractions is equivalent to 0.4 ?

Matriculation Level 01/08/2025 (Shift - 3)

(a) $\frac{2}{3}$ (b) $\frac{3}{5}$ (c) $\frac{1}{4}$ (d) $\frac{2}{5}$

Q.123. Simplify:

$$\sqrt{18} + \sqrt{50}$$

Higher Secondary 25/07/2025 (Shift - 2)

(a) $\sqrt{68}$ (b) $3\sqrt{2} + 5\sqrt{2}$ (c) $8\sqrt{2}$ (d) $7\sqrt{2}$

Q.124. The value of $\sqrt{(7 + 4\sqrt{3})}$ is:

Higher Secondary 28/07/2025 (Shift - 2)

(a) $\sqrt{3} + \sqrt{1}$ (b) $\sqrt{3} + 2$

(c) $\sqrt{6} + \sqrt{1}$ (d) $\sqrt{3} + \sqrt{1}$

Q.125. The square root of $9 + 4\sqrt{5}$ is:

Higher Secondary 29/07/2025 (Shift - 2)

(a) $2 + \sqrt{5}$ (b) $\sqrt{5} - 2$ (c) $3 + \sqrt{5}$ (d) $3 - \sqrt{5}$

Q.126. Simplification:

$$(\sqrt{5} + \sqrt{7})^2$$

Higher Secondary 29/08/2025 (Shift - 2)

(a) $12 + \sqrt{35}$ (b) $12 + 2\sqrt{35}$

(c) $12 + 2\sqrt{5}$ (d) $12 + 2\sqrt{7}$

Q.127. What is the value of $(0.96 \div 8) \div (1.30 - 0.34) = ?$

Graduate Level 24/07/2025 (Shift - 3)

(a) $\frac{1}{6}$ (b) $\frac{1}{8}$ (c) $\frac{1}{4}$ (d) $\frac{3}{8}$

Q.128. Find the value of:

$$\sqrt{8 + 2\sqrt{15}} + \sqrt{8 - 2\sqrt{15}}$$

Graduate Level 24/07/2025 (Shift - 3)

(a) $3\sqrt{3}$ (b) $2\sqrt{5}$ (c) $3\sqrt{2}$ (d) $\sqrt{15}$

Q.129. Evaluate:

$$(\frac{1}{21}) + (\frac{1}{42}) + (\frac{1}{63}) + (\frac{1}{28})$$

Graduate Level 25/07/2025 (Shift - 3)

(a) 0.01231 (b) 0.01002



(c) 0.12301 (d) 0.01356

Q.130. Evaluate:

$$\left(\frac{1}{21}\right) + \left(\frac{1}{28}\right) + \left(\frac{1}{35}\right) + \left(\frac{1}{42}\right)$$

Graduate Level 26/07/2025 (Shift - 3)

(a) 0.1928 (b) 0.0856
(c) 0.1357 (d) 0.0775**Q.131.** Simplify:

$$\left(5\frac{1}{2} + 3.75\right) \times \left(\frac{3}{8} - 0.125\right) \div \left(1\frac{2}{5} + \frac{4}{10}\right)$$

Graduate Level 28/07/2025 (Shift - 3)

(a) $\frac{185}{144}$ (b) $\frac{175}{144}$ (c) $\frac{165}{144}$ (d) $\frac{145}{144}$ **Q.132.** Simplify: $\sqrt{108} + \sqrt{12} - \sqrt{75}$

Graduate Level 29/07/2025 (Shift - 3)

(a) $5\sqrt{3}$ (b) $7\sqrt{3}$ (c) $3\sqrt{3}$ (d) $6\sqrt{3}$ **Q.133.** Simplify:

$$(4\sqrt{7} + 3\sqrt{28}) - (\sqrt{63} + 5\sqrt{7})$$

Graduate Level 31/07/2025 (Shift - 3)

(a) 0 (b) $2\sqrt{7}$ (c) $5\sqrt{7}$ (d) $-3\sqrt{5}$ **Q.134.** Simplify $(\sqrt{2} + \sqrt{7})^2$.

Graduate Level 02/08/2025 (Shift - 3)

(a) $3 - 2\sqrt{15}$ (b) $8 - 2\sqrt{14}$
(c) $2 + 2\sqrt{15}$ (d) $9 + 2\sqrt{14}$ **Q.135.** Find the value of:

$$\left(\frac{1}{7 - \sqrt{6}}\right) + \left(\frac{1}{7 + \sqrt{6}}\right) - \left(\frac{14}{49 - 6}\right)$$

Graduate Level 02/08/2025 (Shift - 3)

(a) 0 (b) 14 (c) $\sqrt{6}$ (d) 7**Q.136.** Find the value of

$$0.2 + 0.22 + 0.222$$

Graduate Level 29/08/2025 (Shift - 1)

(a) 0.642 (b) 0.664 (c) 0.662 (d) 0.644

eduquity-based pattern (ebp) Questions

Q.137. The value of $5 + 24 \times 8 \div 8$ of $8 - 8 \div 8 \times 4$ of 4 is equal to:

(a) 8 (b) 9 (c) -8 (d) -9

Q.138. Simplify $(2z - 5y)^2 + (5z + 2y)^2 - 25z^2$.(a) $19y^2 - 4z^2$ (b) $29y^2 + 4z^2$
(c) $29y^2 - 4z^2$ (d) $19y^2 + 4z^2$ **Q.139.** The lowest terms of $\frac{221}{488}$ is:(a) $\frac{121}{488}$ (b) $\frac{221}{488}$ (c) $\frac{121}{288}$ (d) $\frac{221}{288}$ **Q.140.** Simplify $\frac{(x^2 - 9)}{x + 3}$ (a) $x + 3$ (b) $x - 3$ (c) $\sqrt{x^2 - 9}$ (d) $x - 9$ **Q.141.** Simplify

$$\frac{(8.3)^3 + (9.2)^3 + (6.1)^3 - 3 \times 8.3 \times 9.2 \times 6.1}{(8.3)^2 + (9.2)^2 + (6.1)^2 - 8.3 \times 9.2 - 9.2 \times 6.1 - 6.1 \times 8.3}$$

(a) 30.2 (b) 23.6 (c) 28.7 (d) 25.5

Q.142. Simplify

$$\frac{0.58 \times 0.58 \times 0.58 - 0.001}{0.58 \times 0.58 + 0.058 + 0.01'}$$

(a) 0.48 (b) 0.28 (c) 0.37 (d) 0.42

Q.143. If $45 \div 3 + 4 \times A - 48 \div 24 + 4 = 1$, then the value of A is:

(a) -2 (b) 4 (c) 2 (d) -4

Q.144. Simplify $\frac{(4.2)^3 - (1.2)^3}{(4.2)^2 + 5.04 + (1.2)^2}$

(a) 2 (b) 3 (c) 1 (d) 5

Q.145. The value of the expression $(1.55^2 + 0.55^2 + 1.55 \times 0.55)$ is equal to:(a) $1.55^2 - 0.55^2$ (b) $1.55^3 - 0.55^3$
(c) $1.55^3 + 0.55^3$ (d) $1.55^2 + 0.55^2$ **Q.146.** Solve the following:

$$[(2.5 + 1.5)^2 \div 4 \text{ of } \frac{1}{2} \times 8 - 1]$$

(a) 0 (b) 81 (c) 27.1 (d) 63

Q.147. Simplify the following expression.

$$\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x-y)(y-z)(z-x)}$$

(a) 3 (b) 1 (c) 0 (d) 2

Q.148. Simplify the following expression.

$$(g-h)^2 - (g+h)^2 + 4gh$$

(a) $4gh$ (b) $-4gh$ (c) 4 (d) 0**Q.149.** Simplify

$$\frac{1.5 \times 1.5 + 2.5 \times 2.5 + 3.5 \times 3.5 + 2 \times 1.5 \times 2.5 + 2 \times 2.5 \times 3.5 + 2 \times 1.5 \times 3.5}{1.5 + 2.5 + 3.5}$$

(a) 9.5 (b) 6.5 (c) 8.5 (d) 7.5

Q.150. Simplify the following:

$$[(5\frac{1}{2} - 3\frac{1}{3}) \text{ of } 1\frac{5}{13} + 1 - 2 \times 3\frac{5}{2} \div 2\frac{3}{4}]$$

(a) 0 (b) $\frac{11}{2}$ (c) -2 (d) $3\frac{1}{2}$ **Q.151.** Simplify the given expression.

$$\frac{(4x^2 + 9y^2 - 12xy)}{(3y - 2x)}$$

(a) $2y - 3x$ (b) $3y - 2x$ (c) $2x - 3y$ (d) $3x - 2y$ **Q.152.** Simplify the following expression

$$25 \div 5 + 3 \text{ of } 2 - 4 + 6 \text{ of } 3 \div 2 - 8 + 4$$

(a) 16 (b) 4 (c) 8 (d) 12

Q.153. Simplify the following expression.

$$0.07 \times 0.28 \div 0.04 + 0.64 - 1.64 \div 0.04$$

(a) 38.56 (b) -39.87 (c) -46.48 (d) 36.78

Q.154. Find the value of

$$11 - \{(3 \div 4) \text{ of } 8 - 4 - 8\}$$

(a) 2 (b) 3 (c) 1 (d) -1

Q.155. Simplify the following expression.

$$133 \div 19 \times 3 - 48 \times 2 \div 24 + 841 \div \{[45 - 16]\}$$

(a) 46 (b) 21 (c) 12 (d) 42

Q.156. $9 \div [1 + \{4 \times (\frac{5}{6} - \frac{1}{3} + \frac{1}{2})\}]$

is equal to:

(a) 1 (b) $\frac{9}{5}$ (c) 0 (d) 9**Q.157.** The value of N in the equation

$$9 \times N \text{ of } 4 \div 4 - 15 \times 4 = 12 \text{ is:}$$

(a) 9 (b) 0 (c) 12 (d) 8

Q.158. If $\frac{3(16^3 - 6^3)}{16^2 + 6^2 + Q} = 30$, then find the

value of Q.

(a) 96 (b) 98 (c) 112 (d) 108

Q.159. Simplify the following expression.

$$(60 + 64 \div 4 \text{ of } 4) \times \{1000 \div ((10 \text{ of } (3 + 2)) \times 2 \text{ of } 5)\} - 28 + (100 \times 3 \text{ of } 3)$$

(a) 600 (b) 1000 (c) 100 (d) 500

Q.160. Simplify $626 \times 2912 \div 8138$

(a) 285 (b) 224 (c) 364 (d) 310

Q.161. Simplify $\frac{4x^2 - 49y^2}{2x + 7y} + 7y - 2x$ (a) $4x - 14y$ (b) $7y$ (c) $4x$ (d) 0**Q.162.** The value of

$$\left(\frac{7 + 8 \times 7 \div 8 + 7 \div 7 \div 7}{3 + 3 \times 3 \div 3 + 5 \div 5 \div 5}\right) \text{ is:}$$

(a) $1\frac{81}{217}$ (b) $2\frac{61}{217}$ (c) $3\frac{27}{217}$ (d) $1\frac{31}{217}$ **Q.163.** What is the value of the following expression?

$$\frac{4}{7} \times 4 \frac{1}{2} \div 5 \frac{1}{3} \text{ of } 2 \frac{1}{2} - (7 \frac{7}{8} \div 5 \frac{1}{9} \text{ of } 9 \frac{9}{20}) + \frac{1}{2}$$

(a) $\frac{1759}{3250}$ (b) $\frac{1113}{856}$ (c) $\frac{853}{1610}$ (d) $\frac{1093}{111}$ **Q.164.** Simplify the following expression.

$$(3 - 3 \times 3 + 3 \div 3 + 3 \times 5) \times 2 \text{ of } 5 + (2 + 2 \div 2 + 2 \times 2 - 2)$$

(a) 102 (b) 100 (c) 105 (d) 106

Q.165. Simplify:

$$\{(3^3 + 2^3) \times \frac{22}{7}\} \times \{(6^2 + 3^2 - 4^2) \div (14 \frac{1}{2})\}$$

(a) 220 (b) 145 (c) 110 (d) 198

Q.166. Simplify the following expression.

$$[40 - 48 \div 4 \text{ of } 3] \div \{(6 + 4) - 4 \text{ of } 4 + 4 + (6 - 4) + (4 \times 4) \div 4\} \times 3$$

(a) 30 (b) 26 (c) 25 (d) 27

Q.167. Simplify the given expression.

$$8 - 5 \div 2^3 \times 3 + (7 \div 7 \times 7) \div 7 + (3 \times 6 \div \frac{1}{3}) \times \frac{1}{3^3}$$

(a) $\frac{73}{9}$ (b) $\frac{73}{27}$ (c) $\frac{73}{21}$ (d) $\frac{73}{8}$ **Q.168.** The value of $10 + 3 \frac{3}{5} \div [2 \frac{2}{3} \div$

$$\{7\frac{1}{6} - (2\frac{1}{2} + \frac{2}{3})\}]$$
 is equal to:



(a) $\frac{27}{55}$ (b) $\frac{77}{5}$ (c) $\frac{55}{27}$ (d) $\frac{5}{77}$

Q.169. Simplify:

$$3\frac{4}{7} - [16\frac{1}{2} \div \{6^2 - (7 - 2\frac{1}{2})\}]$$

(a) $\frac{25}{11}$ (b) $\frac{64}{21}$ (c) $\frac{64}{11}$ (d) $\frac{25}{21}$

Q.170. Simplify $\frac{\frac{1}{2} \div \frac{3}{4} \text{ of } \frac{5}{6}}{\frac{1}{2} \div \frac{3}{4} \times \frac{5}{6}} + \frac{2+2 \times 2}{2 \div 2 \times 2}$

(a) $\frac{111}{25}$ (b) $\frac{121}{25}$ (c) $\frac{75}{36}$ (d) 4

Q.171. Simplify the following expression.

$$4 \times 7 - 3\{7 \times 4 \div (6 \times 2)\}$$

(a) 12 (b) 19 (c) 21 (d) 4

Q.172. What is the value of the following expression?

$$420 \div [\frac{4}{9} \times (1 + \frac{4}{5}) \times 5] + [\frac{5}{7} \text{ of } \frac{21}{25} \div \frac{3}{4} \text{ of } \frac{2}{5}]$$

(a) 127 (b) 139 (c) 107 (d) 89

Q.173. Solve

$$\frac{2 \times 2 + 2}{2 \div 2 \times 2} \div \frac{\frac{1}{2} \div \frac{1}{2} \text{ of } \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \text{ of } \frac{1}{2}}$$

(a) $4\frac{1}{2}$ (b) $2\frac{1}{8}$ (c) $1\frac{1}{8}$ (d) $3\frac{1}{8}$

Q.174. What is the value of the following expression?

$$(24 \div 3 \text{ of } \frac{1}{6}) \times (\frac{4}{5} \div \frac{9}{8} \times \frac{3}{5}) \div (\frac{4}{7} \div \frac{3}{14} \text{ of } \frac{3}{14})$$

(a) $\frac{379}{179}$ (b) $\frac{579}{459}$ (c) $\frac{363}{553}$ (d) $\frac{288}{175}$

Q.175. The value of

$$\frac{5 + 7 \times 5 \div 8 \text{ of } 5 + 3 \div 3 \times 5}{6 \div 6 \text{ of } 4 + 9 \times 9 \div 3 \times 2 \times 2} \text{ is :}$$

(a) $\frac{67}{859}$ (b) $\frac{87}{866}$ (c) $\frac{97}{881}$ (d) $\frac{77}{861}$

Q.176. Simplify the following expression.

$$11\frac{1}{4} \div 3\frac{3}{5} \times 6\frac{2}{3} + \frac{1}{6} = \text{_____}$$

(a) 19 (b) 21 (c) 23 (d) 25

Q.177. The value of $[3^3 \{(40 \times 5) + (7 \times 6)\} \div 11] + (2^3 \times 15 \div 24)$ is:

(a) 559 (b) 595 (c) 599 (d) 959

Q.178. If '÷' means 'x', 'x' means '÷', '+' means '-', '-' means '+', then $47 \div (10 \times 5) + (4 - 5) - 6$ is:

(a) 100 (b) 59 (c) 79 (d) 91

Q.179. What is the value of

$$\frac{[\frac{1}{4} \div \frac{1}{4} \times \frac{1}{4}] \times \frac{1}{4}}{[\frac{1}{8} + \frac{1}{8} \times \frac{1}{8}]} + (1 - \frac{4}{9})?$$

(a) 1 (b) 0 (c) 3 (d) 2

Q.180. Simplify the following expression.

$$3\frac{1}{5} \text{ of } 2\frac{3}{4} \div \frac{11}{3} + (2\frac{1}{5} \div 1\frac{2}{5} \text{ of } 1\frac{3}{7} - 2\frac{1}{3})$$

(a) $4\frac{1}{6}$ (b) $2\frac{1}{6}$ (c) $3\frac{1}{6}$ (d) $1\frac{1}{6}$

Q.181. Find the value of $[18 - 3 \times \{3 - 3 \times$

$$(3 \div \frac{1}{3} - 3) + 3 \div \frac{1}{3} + 3\} - 4 \times 2].$$

(a) 18 (b) 19 (c) 12 (d) 15

Q.182. Find the value of $12 + 24 \div 2 - 7 \times 30 \div 6 + (5 + 4) \times 8 + 8 \div 8 - 8$.

(a) 45 (b) 85 (c) 54 (d) 58

Q.183. The possible value of x satisfying

$$\frac{(143 \div 13 \times x - 3 \times 3)}{(6^2 - 7 \times 5 + x^2)} = 1 \text{ is :}$$

(a) 1 (b) 3 (c) 2 (d) 4

Q.184. Find the value of

$$12 + 3\frac{2}{3} \div [1\frac{3}{4} \div \{5\frac{1}{4} \times (3\frac{3}{5} \div 4\frac{1}{2})\}]$$

(a) 5 (b) 3 (c) 1 (d) 2

Q.185. Simplify the following expression:

$$25 \times 25 + 9 \times 9 + 6 \times 6 + 2 \times 25 \times 9 + 2 \times 9 \times 6 + 2 \times 25 \times 6$$

(a) 900 (b) 1200 (c) 1600 (d) 2500

Answer Key :-

1.(c)	2.(b)	3.(a)	4.(a)
5.(c)	6.(b)	7.(a)	8.(a)
9.(b)	10.(a)	11.(a)	12.(c)
13.(b)	14.(d)	15.(a)	16.(b)
17.(d)	18.(c)	19.(c)	20.(d)
21.(d)	22.(d)	23.(a)	24.(c)
25.(a)	26.(c)	27.(a)	28.(b)
29.(d)	30.(a)	31.(d)	32.(d)
33.(a)	34.(b)	35.(c)	36.(c)
37.(b)	38.(a)	39.(c)	40.(a)
41.(c)	42.(a)	43.(a)	44.(c)
45.(d)	46.(b)	47.(d)	48.(b)
49.(a)	50.(b)	51.(d)	52.(a)
53.(c)	54.(b)	55.(b)	56.(a)
57.(d)	58.(a)	59.(c)	60.(a)
61.(c)	62.(b)	63.(a)	64.(b)
65.(b)	66.(b)	67.(a)	68.(a)
69.(d)	70.(d)	71.(a)	72.(d)
73.(c)	74.(a)	75.(c)	76.(c)
77.(d)	78.(c)	79.(d)	80.(b)
81.(a)	82.(a)	83.(b)	84.(a)

85.(a)	86.(b)	87.(d)	88.(d)
89.(a)	90.(a)	91.(a)	92.(d)
93.(c)	94.(a)	95.(a)	96.(b)
97.(a)	98.(b)	99.(a)	100.(b)
101.(a)	102.(b)	103.(b)	104.(c)
105.(d)	106.(b)	107.(b)	108.(a)
109.(a)	110.(a)	111.(a)	112.(b)
113.(b)	114.(a)	115.(b)	116.(b)
117.(a)	118.(b)	119.(a)	120.(b)
121.(a)	122.(d)	123.(c)	124.(b)
125.(a)	126.(b)	127.(b)	128.(b)
129.(c)	130.(c)	131.(a)	132.(c)
133.(b)	134.(d)	135.(a)	136.(a)
137.(c)	138.(b)	139.(b)	140.(b)
141.(b)	142.(a)	143.(d)	144.(b)
145.(b)	146.(d)	147.(a)	148.(d)
149.(d)	150.(a)	151.(b)	152.(d)
153.(b)	154.(c)	155.(a)	156.(d)
157.(d)	158.(a)	159.(b)	160.(b)
161.(d)	162.(b)	163.(c)	164.(c)
165.(a)	166.(d)	167.(d)	168.(b)
169.(b)	170.(a)	171.(c)	172.(c)
173.(c)	174.(d)	175.(b)	176.(b)
177.(c)	178.(d)	179.(a)	180.(d)
181.(b)	182.(c)	183.(a)	184.(b)
185.(c)			

Solutions :-

Sol.1.(c) Given :-

$$\sqrt{20 + \sqrt{20 + \sqrt{20 + \dots}}}$$

In this type question we check prime factor, $20 = 4 \times 5$

If a positive series is given in the question, then the answer will be of larger factor. And if a negative series is given in the question, then the answer will be of the smallest factor.

Hence, the answer is 5.

Sol.2.(b) Given , $\sqrt{7} + \sqrt{3}, \sqrt{6} + \sqrt{3}$

$$\text{Now , } (\sqrt{7} + \sqrt{3})^2 = 7 + 3 + 2\sqrt{7} \times \sqrt{3} = 10 + 2\sqrt{21} = 10 + 9.16 = 19.16$$

$$\text{Now , } (\sqrt{6} + \sqrt{3})^2 = 6 + 3 + 2\sqrt{6} \times \sqrt{3} = 9 + 2\sqrt{18} = 9 + 8.48 = 17.48$$

Hence , conditions I and II are incorrect .

$$\text{Sol.3.(a) } (\frac{1}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}}) = 2 + \sqrt{3}$$

Now,

$$(2 + \sqrt{3}) + (2 + \sqrt{3}) - 4 - 2\sqrt{3} = 0$$



Sol.4.(a)

$$(\sqrt{576} - \sqrt{144}) \times (\sqrt{64} + \sqrt{225})$$

$$= (24 - 12) \times (8 + 15) = 12 \times 23 = 276$$

Sol.5.(c)

Express the decimal as a fraction:

$$0.365 = \frac{365 \div 5}{1000 \div 5} = \frac{73}{200}$$

$$\text{Sol.6.(b)} \quad \frac{(25)^4 + 1250 + 1 - (25)^2}{(25)^2 + 25 + 1}$$

$$= \frac{(25)^4 - (25)^2 + 1250 + 1}{(25)^2 + 25 + 1}$$

$$= \frac{[(25)^2]^2 + 2 \times 25^2 \times 1 + (1)^2 - (25)^2}{(25)^2 + 25 + 1}$$

$$= \frac{[(25)^2 + 1]^2 - (25)^2}{(25)^2 + 25 + 1}$$

$$= \frac{[(25^2 + 1 + 25)(25^2 + 1 - 25)]}{(25)^2 + 25 + 1}$$

$$= (25^2 + 1 - 25) = 625 - 24 = 601$$

$$\text{Sol.7.(a)} \quad \frac{25a^2 - 10ab - 48b^2}{5a + 6b} \times (5a + 8b)$$

$$\frac{25a^2 - 40ab + 30ab - 48b^2}{5a + 6b} \times (5a + 8b)$$

$$\frac{5a(5a - 8b) + 6b(5b - 8b)}{5a + 6b} \times (5a + 8b)$$

$$\frac{(5a - 8b)(5a + 6b)}{5a + 6b} \times (5a + 8b)$$

$$(5a - 8b) \times (5a + 8b)$$

$$(5a)^2 - (8b)^2 = 25a^2 - 64b^2$$

Sol.8.(a)

$$= \frac{(x^2 - 9) \times (9x^2 - 1)}{(x - \frac{1}{3}) \times (1 - \frac{x}{3})} \times \frac{9}{(x + \frac{1}{3})(1 + \frac{x}{3})}$$

$$= \frac{(x + 3)(x - 3)(3x + 1)(3x - 1)}{(3x - 1)(3 - x)} \times$$

$$\frac{9}{(3x + 1)(3 + x)} = -729$$

Sol.9.(b)

$$A = \frac{58^2 - 25^2}{46^2 - 37^2} = \frac{(58 - 25)(58 + 25)}{(46 - 37)(46 + 37)}$$

$$= \frac{33 \times 83}{9 \times 83} = \frac{11}{3}$$

$$B = \frac{26^2 - 15^2}{56^2 - 15^2} = \frac{(26 - 15)(26 + 15)}{(56 - 15)(56 + 15)}$$

$$= \frac{11 \times 41}{41 \times 71} = \frac{11}{71}$$

$$\text{Then, } \frac{1}{B} - \frac{20}{A} = \frac{71}{11} - \frac{20 \times 3}{11} = \frac{11}{11} = 1$$

Sol.10.(a)

$$\Rightarrow [(2\frac{7}{9})^2]^{\frac{3}{5}} = [(\frac{5}{3})^2]^{\frac{5}{2}}]^{\frac{3}{5}} = (\frac{5}{3})^3$$

$$\Rightarrow [(1\frac{2}{3})^5]^{\frac{3}{5}} = (\frac{5}{3})^3$$

∴ both the expressions are equal

Sol.11.(a)

$$\frac{24\sqrt{45} - 16\sqrt{20} + \sqrt{245} - 47\sqrt{5}}{3\frac{1}{5} + 1\frac{2}{5} - 2\frac{1}{6} - 2\frac{1}{5} - 5\frac{2}{5} + 3\frac{1}{6}}$$

$$= \frac{72\sqrt{5} - 32\sqrt{5} + 7\sqrt{5} - 47\sqrt{5}}{\frac{16}{5} + \frac{7}{5} - \frac{13}{6} - \frac{11}{5} - \frac{27}{5} + \frac{19}{6}}$$

$$= \frac{79\sqrt{5} - 79\sqrt{5}}{\frac{16}{5} + \frac{7}{5} - \frac{13}{6} - \frac{11}{5} - \frac{27}{5} + \frac{19}{6}} = 0$$

Sol.12.(c)

$$\frac{(3.321)^3 + (2.681)^3 + (1.245)^3 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

$$\frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} = (a + b + c)$$

Here, $3.321 + 2.681 + 1.245 = 7.247$

$$\text{Sol.13.(b)} \quad 0.01 + 0.68 \times 0.68 + 0.22 \times 0.22 + 0.44 + 0.136 + 0.44 \times 0.68$$

$$= (0.68 + 0.22)^2 + 0.01 + 0.044 + 0.136$$

$$0.81 + 0.19 = 1$$

$$\text{Sol.14.(d)} \quad (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\frac{0.09 \times 0.09 + 0.04 \times 0.04 + 0.16 \times 0.16 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.3 \times 0.3 + 0.2 \times 0.2 + 0.4 \times 0.4}$$

$$= \frac{(0.09 + 0.04 + 0.16)^2}{0.09 + 0.04 + 0.16}$$

$$= 0.09 + 0.04 + 0.16 = 0.29$$

$$\text{Sol.15.(a)} \quad \frac{4^{a+4} - 5 \times 4^{a+2}}{15 \times 4^a - 2^2 \times 4^a}$$

$$= \frac{4^a(4^4 - 5 \times 4^2)}{4^a(15 - 2^2)} = \frac{(256 - 80)}{11} = 16$$

Sol.16.(b)

$$(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)$$

Multiply and divide by $(2^2 - 1)$ in the given equation, we get

$$\frac{(2^2 - 1)(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{(2^2 - 1)}$$

$$= \frac{(2^4 - 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{3}$$

$$= \frac{(2^8 - 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{3}$$

$$= \frac{(2^{256} - 1)}{3}$$

$$\text{Sol.17.(d)} \quad \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{110} < \frac{5}{6}$$

$$= (1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + \dots$$

$$(\frac{1}{10} - \frac{1}{11}) < \frac{5}{6} = 1 - \frac{1}{11} = \frac{10}{11} > \frac{5}{6}$$

Hence, the first statement is wrong.

$$\text{now, } \frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{143} > \frac{7}{13}$$

$$= \frac{1}{2}[(1 - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{5}) + (\frac{1}{5} - \frac{1}{7})$$

$$+ \dots + (\frac{1}{11} - \frac{1}{13})] > \frac{7}{13} = \frac{1}{2}[(1 - \frac{1}{13})$$

$$= \frac{6}{13} < \frac{7}{13}$$

Hence, the second statement is also wrong.

Sol.18.(c)

$$[\frac{5}{8} - \{\frac{3}{8} - (\frac{5}{8} - \frac{3}{8})\}] \text{ of } 8.8 - 1.2$$

$$4\frac{1}{6} \div 2.5 \times 2 \div \frac{1}{6} \text{ of } 60 + (\frac{3}{4} - \frac{3}{8})$$

$$[\frac{5}{8} - \{\frac{3}{8} - \frac{1}{4}\}] \text{ of } 8.8 - 1.2$$

$$= \frac{\frac{25}{6} \div 2.5 \times 2 \div 10 + \frac{3}{8}}$$

$$[\frac{5}{8} - \frac{1}{8}] \text{ of } 8.8 - 1.2$$

$$= \frac{\frac{25}{6} \div \frac{5}{2} \times 2 \div 10 + \frac{3}{8}}$$

$$= \frac{\frac{1}{2} \text{ of } 8.8 - 1.2}{\frac{5}{3} \times \frac{1}{5} + \frac{3}{8}} = \frac{4.4 - 1.2}{\frac{1}{3} + \frac{3}{8}}$$

$$= \frac{3.2}{\frac{17}{24}} = 4\frac{44}{85}$$

Sol.19.(c)

$$24 \div \frac{3}{8} \text{ of } (8 + 2 \times \sqrt{7-3}) + [\frac{2}{11} \div \frac{4}{55} - \{\frac{5}{8} + \frac{6}{16}\}]$$

$$32 \div \sqrt{15-7} + 75 \div (6 + 15 \div 3 + 4)$$

$$24 \div \frac{3}{8} \text{ of } (8 + 2 \times 4) + [\frac{5}{2} - 1]$$

$$= \frac{32 \div 8 + 75 \div (6 + 5 + 4)}{24 \div \frac{3}{8} \text{ of } 16 + 1.5} = \frac{4 + 1.5}{9}$$

$$= \frac{5.5}{9} = \frac{11}{18}$$

Sol.20.(d)

$$\frac{(6.9)^3 + (2.7)^3 + (5.8)^3 - 3 \times 6.9 \times 2.7 \times 5.8}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)}$$

$$\frac{(6.9 + 2.7 + 5.8)[(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9 \times 2.7) - (2.7 \times 5.8) - (5.8 \times 6.9)]}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)}$$

$$= 6.9 + 2.7 + 5.8 = 15.4$$

Sol.21.(d)

$$9\frac{1}{5} \div \{(16\frac{1}{5} \div 12\frac{2}{3} - 9\frac{2}{3} + 17\frac{1}{2}$$

$$\text{of } 3\frac{1}{21})\}$$

$$= \frac{46}{5} \div \{\frac{81}{5} \div (\frac{38}{3} - \frac{29}{3}) + \frac{35}{2}\}$$

$$\text{of } \frac{64}{21}$$

$$= \frac{46}{5} \div \{\frac{81}{5} \div 3 + \frac{160}{3}\}$$

$$= \frac{46}{5} \div \{\frac{27}{5} + \frac{160}{3}\}$$

$$= \frac{46}{5} \div \frac{881}{15} \Rightarrow \frac{46}{5} \times \frac{15}{881} \Rightarrow \frac{138}{881}$$

Sol.22.(d)

$$\frac{1}{4} + \frac{[(20.35)^2 - (8.35)^2] \times 0.0175}{(1.05)^2 + (1.05)(27.65)}$$

$$= \frac{1}{4} + \frac{[(20.35 + 8.35)(20.35 - 8.35)] \times 0.0175}{(1.05)\{(1.05) + (27.65)\}}$$

$$= \frac{1}{4} + \frac{[28.7 \times 12] \times 0.0175}{(1.05)(28.7)}$$

$$= \frac{1}{4} + \frac{12 \times 175}{105 \times 100} = \frac{1}{4} + \frac{12 \times 5}{3 \times 100}$$



$$= \frac{1}{4} + \frac{1}{5} = \frac{9}{20}$$

Sol.23.(a)

$$\frac{119 - [48 \div 6 - 7 \{5 \times 12 \div 3 - (15 - 3 - 8)\}]}{24 \div 3 - 9 \text{ of } 3 + (52 - 8 - 4) \div 6}$$

$$= \frac{119 - [8 - 7 \{5 \times 4 - (15 + 5)\}]}{8 - 27 + (52 - 4) \div 6}$$

$$= \frac{119 - [8 - 7 \{20 - (20)\}]}{8 - 27 + (48) \div 6}$$

$$= \frac{119 - [8 - 7 \{0\}]}{8 - 27 + 8}$$

$$= \frac{119 - [8]}{-11} = -\frac{111}{11}$$

Sol.24.(c) $\frac{1}{4 - \sqrt{8}}$ (on rationalizing)

$$= \frac{1}{4 - \sqrt{8}} \times \frac{4 + \sqrt{8}}{4 + \sqrt{8}} = \frac{4 + \sqrt{8}}{8} \text{ so,}$$

$$= \frac{4 + \sqrt{8}}{8} + \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} - \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}}$$

$$= \frac{4 + \sqrt{8}}{8} + \frac{(3 + 2\sqrt{2})^2 - (3 - 2\sqrt{2})^2}{(3)^2 - (2\sqrt{2})^2}$$

$$= \frac{4 + \sqrt{8}}{8} + \frac{24\sqrt{2}}{9 - 8} \Rightarrow \frac{1}{2} + \frac{\sqrt{2}}{4} + 24\sqrt{2}$$

Since, $\frac{1}{2} + \frac{97}{4}\sqrt{2} = a + b\sqrt{2}$

On comparing $a = \frac{1}{2}$, $b = \frac{97}{4}$

$$3a + 4b = 3 \times \frac{1}{2} + 4 \times \frac{97}{4} = 98\frac{1}{2}$$

Sol.25.(a) After checking option one by one option (a) satisfies.

$$4 \times 30 - 15 \div 5 + 12 = 21$$

$$\Rightarrow 4 \times 30 \div 5 - 15 + 12 = 21$$

$$\Rightarrow 4 \times 6 - 15 + 12 = 21$$

$$\Rightarrow 24 - 15 + 12 = 21 \Rightarrow 9 + 12 = 21$$

$$\Rightarrow 21 = 21$$

Sol.26.(c) $\frac{4}{1 + \sqrt{2} + \sqrt{3}}$

$$= a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$$

$$= \frac{4}{1 + \sqrt{2} + \sqrt{3}} \times \frac{\sqrt{3} + 1 - \sqrt{2}}{\sqrt{3} + 1 - \sqrt{2}}$$

$$= (a + b)(a - b) = a^2 - b^2$$

$$\frac{4\sqrt{3} + 4 - 4\sqrt{2}}{(\sqrt{3} + 1)^2 - (\sqrt{2})^2} = \frac{4\sqrt{3} + 4 - 4\sqrt{2}}{2\sqrt{3} + 2}$$

$$= \frac{2\sqrt{3} + 2 - 2\sqrt{2}}{\sqrt{3} + 1} \times \frac{\sqrt{3} - 1}{\sqrt{3} - 1}$$

$$= \frac{(2\sqrt{3} + 2 - 2\sqrt{2})(\sqrt{3} - 1)}{3 - 1}$$

$$= \frac{(2\sqrt{3} + 2 - 2\sqrt{2})(\sqrt{3} - 1)}{2}$$

$$= (\sqrt{3} + 1 - \sqrt{2})(\sqrt{3} - 1)$$

$$= 3 + \sqrt{3} - \sqrt{6} - \sqrt{3} - 1 + \sqrt{2}$$

$$= 2 + \sqrt{2} - \sqrt{6} = a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6}$$

$$a = 2; b = 1; c = 0; d = 1$$

$$a + b + c + d = 2 + 1 + 0 + 1 = 4$$

Sol.27.(a)

$$\left[\left(\frac{2}{3}\right)^3\right]^{2x+3} \cdot \left[\left(\frac{2}{3}\right)^3\right]^{\frac{-3}{4}} = \left[\left(\frac{2}{3}\right)^3\right]^{3x+7} \cdot \left[\left(\frac{2}{3}\right)^3\right]^{\frac{-6}{5}}$$

$$\Rightarrow \left[\left(\frac{2}{3}\right)^3\right]^{(2x+3) \cdot \frac{3}{4}} = \left[\left(\frac{2}{3}\right)^3\right]^{(3x+7) \cdot \frac{6}{5}}$$

$$\Rightarrow \left[\left(\frac{2}{3}\right)^3\right]^{\frac{3(2x+3)}{4}} = \left[\left(\frac{2}{3}\right)^3\right]^{\frac{6(3x+7)}{5}}$$

$$\Rightarrow \left[\left(\frac{2}{3}\right)^3\right]^{\frac{6x+9}{4}} = \left[\left(\frac{2}{3}\right)^3\right]^{\frac{18x+42}{5}}$$

$$\Rightarrow \left(\frac{2}{3}\right)^{\frac{3(6x+9)}{4}} = \left(\frac{2}{3}\right)^{\frac{2(18x+42)}{3 \times 5}}$$

$$= \left[\left(\frac{2}{3}\right)^{\frac{2(18x+42)}{3 \times 5}}\right] = \left[\left(\frac{2}{3}\right)^{\frac{18x+27}{4}}\right]$$

$$= \frac{36x+84}{15} = \frac{18x+27}{4}$$

$$\Rightarrow 4(36x+84) = 15(18x+27)$$

$$\Rightarrow 144x+336 = 270x-405$$

$$\Rightarrow 126x = 336-405 \Rightarrow 126x = -69$$

$$\Rightarrow x = -\frac{69}{126} = -\frac{23}{42}$$

Now put the value of x in given equation

$$\sqrt{2-42x} = \sqrt{2-42\left(-\frac{23}{42}\right)}$$

$$= \sqrt{2+23} = \sqrt{25} = 5$$

Sol.28.(b) $\frac{42-12 \times 3+8 \div 2+15}{8 \times 2-4+9 \div 3}$

After applying changes as per question we get

$$\frac{42+12 \div 3-8 \times 2-15}{8 \div 2+4-9 \times 3} = -\frac{15}{19}$$

Sol.29.(d) $\frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$

$$\frac{\{(4.6)^2 + (5.4)^2\}^2 - 2\{(4.6)^2(5.4)^2\} + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$= \frac{\{(4.6)^2 + (5.4)^2\}^2 - (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$= \frac{[(4.6)^2 + (5.4)^2 - (24.84)][(4.6)^2 + (5.4)^2 + (24.84)]}{(4.6)^2 + (5.4)^2 + 24.84}$$

$$= (4.6)^2 + (5.4)^2 - 24.84$$

$$= (4.6 + 5.4)^2 - 2(4.6)(5.4) - 24.84$$

$$= 100 - 74.52 = 25.48$$

$$\text{Sol.30.(a)} \frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

$$= \frac{(0.545)(0.081) \times 4 \times 3 \times (0.17)(1.3)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

$$= \frac{3 \times (0.545)(0.324)(0.221)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

Now, since

$$(0.324 + 0.221 - 0.545 = 0)$$

$$\Rightarrow (0.324)^3 + (0.221)^3 - (0.545)^3 =$$

$$- \{3 \times (0.545)(0.324)(0.221)\}$$

....(Algebraic property)

So,

$$\frac{3 \times (0.545)(0.324)(0.221)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$

$$= \frac{3 \times (0.545)(0.324)(0.221)}{-\{3 \times (0.545)(0.324)(0.221)\}} = -1$$

Sol.31.(d)

$$\frac{(0.7)^2 \div 0.14 + (0.6)^2 \div 0.18 + (0.5)^2 \div 0.05}{4(2.5 \text{ of } 4 - 13 \times 0.25 \times 3)}$$

$$= \frac{0.49 \div 0.14 + 0.36 \div 0.18 + 0.25 \div 0.05}{4(10 - 9.75)}$$

$$= \frac{3.5 + 2 + 5}{4 \times 0.25} = 10.5 = \frac{21}{2}$$

Sol.32.(d)

$$x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} \dots + \frac{1}{23.24}$$

$$= \frac{1}{12} - \frac{1}{13} + \frac{1}{13} - \frac{1}{14} + \dots + \frac{1}{23} - \frac{1}{24}$$

$$= \frac{1}{12} - \frac{1}{24} = \frac{1}{24}$$

$$y = \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} \dots + \frac{1}{71.72}$$

$$= \frac{1}{36} - \frac{1}{37} + \frac{1}{37} - \frac{1}{38} + \dots + \frac{1}{71} - \frac{1}{72}$$

$$= \frac{1}{36} - \frac{1}{72} = \frac{1}{72}$$

Therefore, $\frac{x}{y} = \frac{\frac{1}{24}}{\frac{1}{72}} = 3$

Alternate Method :-

$$x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} \dots + \frac{1}{23.24}$$

$$= \frac{1}{13-12} \left(\frac{1}{12} - \frac{1}{24} \right) = \frac{1}{24}$$

$$y = \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} \dots + \frac{1}{71.72}$$

$$= \frac{1}{37-36} \left(\frac{1}{36} - \frac{1}{72} \right) = \frac{1}{72}$$

Therefore, $\frac{x}{y} = \frac{\frac{1}{24}}{\frac{1}{72}} = 3$

Sol.33.(a) $\frac{0.01404}{24^2 + 6^2 - 144}$

$$= \frac{0.01404}{576 + 36 - 144} = \frac{0.01404}{612 - 144}$$

$$= \frac{0.01404}{468} = 3 \times 10^{-5}$$

Sol.34.(b) $\frac{1.0025 + 6.25 \times 10^{-6}}{0.0025 + 0.95}$

$$= \left(\frac{1.00250625}{0.9525} \right) = 1.0525$$

Sol.35.(c) $(3\frac{1}{2} + 4.5) - 1.9$

$$= (3.5 + 4.5) - 1.9 = 8 - 1.9 = 6.1$$

Sol.36.(c)

$$27\frac{1}{4} - \left[\frac{13}{6} \div \left\{ \frac{1}{3} - \left(\frac{1}{2} \times \left(\frac{3}{4} - \frac{1}{4} \right) \right) \right\} \right]$$

$$= \frac{109}{4} - \left[\frac{13}{6} \div \left\{ \frac{1}{3} - \frac{1}{4} \right\} \right]$$

$$= \frac{109}{4} - \left[\frac{13}{6} \times \frac{12}{1} \right]$$

$$= \frac{109}{4} - 26 = \frac{109-104}{4} = 1\frac{1}{4}$$

Sol.37.(b) $32^3 + 15^3 - 35^3 + 250$

$$= 32768 + 3375 - 42875 + 250$$

$$= 36393 - 42875 = -6482$$

Sol.38.(a) Given :- $x = 4 + \frac{1}{5 + \frac{1}{6 + \frac{1}{2}}}$

$$x = 4 + \frac{1}{5 + \frac{1}{\frac{12+1}{2}}} \Rightarrow x = 4 + \frac{1}{5 + \frac{1}{\frac{13}{2}}}$$



$$x = 4 + \frac{1}{5 + \frac{2}{13}} \Rightarrow x = 4 + \frac{1}{\frac{65+2}{13}}$$

$$x = 4 + \frac{13}{67} \Rightarrow x = \frac{4 \times 67 + 13}{67}$$

$$= \frac{268 + 13}{67} = \frac{281}{67}$$

Sol.39.(c)

$$\left(\frac{1}{5 + \sqrt{3}}\right) + \left(\frac{1}{5 - \sqrt{3}}\right) - \left(\frac{10}{87 - 65}\right)$$

$$= \frac{5 - \sqrt{3} + 5 + \sqrt{3}}{5^2 - (\sqrt{3})^2} - \frac{10}{22}$$

$$= \frac{10}{(25 - 3)} - \frac{10}{22} = 0$$

Sol.40.(a)

$$12\left(\frac{1}{2}\right)\% \text{ of } 480 \text{ km} + 66\left(\frac{2}{3}\right)\% \text{ of } 270 \text{ km}$$

$$= \frac{1}{8} \times 480 + \frac{2}{3} \times 270$$

$$= 60 + 180 = 240 \text{ km}$$

$$\text{Sol.41.(c)} 6 - \left[\frac{3}{4} \div \left\{\frac{1}{5} + \left(\frac{1}{2} \times \left(1 - \frac{1}{3}\right)\right)\right\}\right]$$

$$6 - \left[\frac{3}{4} \div \left\{\frac{1}{5} + \left(\frac{1}{2} \times \frac{2}{3}\right)\right\}\right]$$

$$6 - \left[\frac{3}{4} \div \left\{\frac{1}{5} + \frac{1}{3}\right\}\right]$$

$$6 - \left[\frac{3}{4} \div \frac{8}{15}\right] \Rightarrow 6 - \left[\frac{3}{4} \times \frac{15}{8}\right]$$

$$6 - \left[\frac{3}{4} \times \frac{15}{8}\right] \Rightarrow 6 - \frac{45}{32} = \frac{147}{32}$$

$$\text{Sol.42.(a)} 7 + 77 + 777 + 7777 + 77777$$

$$= 7(1 + 11 + 111 + 1111 + 11111)$$

$$= 7 \times 12345 = 86415$$

$$\text{Sol.43.(a)} \left(1\frac{4}{5} + 3.2\right) - 1\frac{1}{4}$$

$$= (1.8 + 3.2) - 1.25 = 3.75$$

Sol.44.(c) Given ,

$$1.2 \times 2\frac{2}{5} \times \frac{3}{6}$$

$$= \frac{12}{10} \times \frac{12}{5} \times \frac{3}{6} = \frac{12 \times 3}{5 \times 5} = \frac{36}{25} = 1.44$$

$$\text{Sol.45.(d)} 0.2 + 0.02 + 0.002 = 0.222$$

$$\text{Sol.46.(b)} 3\frac{1}{4} \div 0.5$$

$$= \frac{13}{4} \times 2 = 6.5$$

$$\text{Sol.47.(d)} \left(\frac{1}{9} \div \frac{1}{3}\right) \div \left(\frac{1}{15} + 1 - \frac{9}{10}\right)$$

$$= \left(\frac{1}{9} \times 3\right) \div \left(\frac{16}{15} - \frac{9}{10}\right)$$

$$= \left(\frac{1}{3}\right) \div \left(\frac{5}{30}\right) = \left(\frac{1}{3}\right) \times 6 = 2$$

$$\text{Sol.48.(b)} \sqrt{98} + \sqrt{32} - \sqrt{50}$$

$$= 7\sqrt{2} + 4\sqrt{2} - 5\sqrt{2} = 6\sqrt{2}$$

Sol.49.(a)

$$\text{Given:-} \left(\frac{8}{5} \div \frac{14}{5}\right) \times \left(\frac{10}{7} \div \frac{4}{3}\right)$$

$$\left(\frac{8}{5} \times \frac{5}{14}\right) \times \left(\frac{10}{7} \times \frac{3}{4}\right) \Rightarrow \frac{4}{7} \times \frac{30}{28} = \frac{120}{196}$$

$$\text{Sol.50.(b) Given:-} \left(3.5 + \frac{3}{8}\right) - \left(\frac{9}{5} \times 1.25\right)$$

$$\left(\frac{7}{2} + \frac{3}{8}\right) - \left(\frac{9}{5} \times \frac{5}{4}\right) \Rightarrow \left(\frac{28+3}{8}\right) - \left(\frac{9}{4}\right)$$

$$\Rightarrow \left(\frac{31}{8} - \frac{9}{4}\right) = \frac{31-18}{8} = 1.625$$

Sol.51.(d) Given:-

$$\left((1.5 + \frac{1}{6}) \div (2.25 - \frac{1}{3})\right) + 0.8$$

$$\left(\frac{3}{2} + \frac{1}{6}\right) \div (2.25 - 0.33) + 0.8$$

$$\left(\frac{10}{6}\right) \div (1.92) + 0.8 = 1.66$$

$$\text{Sol.52.(a) Given :-} \left(\frac{14}{16} \times 4.8\right) \div 0.12$$

$$= 4.2 \div 0.12 = 35$$

$$\text{Sol.53.(c)} 40\% \text{ of } 175 - 37\left(\frac{1}{2}\right)\% \text{ of } 160 +$$

$$10\% \text{ of } 250 \Rightarrow 70 - \frac{3}{8} \times 160 + 25$$

$$= 95 - 60 = 35$$

$$\text{Sol.54.(b)} \sqrt{196} - \sqrt{64} + \sqrt{100}$$

$$= 14 - 8 + 10 = 16$$

$$\text{Sol.55.(b)} \left(3.2 - 1\frac{1}{5}\right) + \left(2\frac{1}{4} \div 0.5\right)$$

$$= (3.2 - 1.2) + \left(\frac{9}{4} \times 2\right) = 2 + 4.5 = 6.5$$

$$\text{Sol.56.(a)} 1 + \frac{1}{3 + \frac{1}{1 + \frac{1}{2}}}$$

$$= 1 + \frac{1}{3 + \frac{1}{\frac{3}{2}}} \Rightarrow 1 + \frac{1}{3 + \frac{2}{3}}$$

$$= 1 + \frac{3}{11} = \frac{14}{11}$$

$$\text{Sol.57.(d)} 2.5\% \text{ of } 100$$

$$= 2.5 \times \frac{1}{100} \times 100 = 2.5$$

Sol.58.(a)

$$\left[\left(\frac{4}{3}\right) + \left(\frac{4}{5} \div \frac{2}{7}\right)\right] \div \left[\left(\frac{3}{2}\right) - \left(\frac{7}{12} \times \frac{3}{4}\right)\right]$$

$$= \left[\left(\frac{4}{3}\right) + \left(\frac{14}{5}\right)\right] \div \left[\left(\frac{3}{2}\right) - \left(\frac{7}{16}\right)\right]$$

$$= \left[\frac{20+42}{15}\right] \div \left[\frac{24-7}{16}\right]$$

$$= \frac{62}{15} \times \frac{16}{17} = \frac{992}{255}$$

Sol.59.(c) Given ,

$$\left(2\frac{1}{2} + 2.6\right) \div 0.75 + \left(1.25 \times \frac{3}{5}\right)$$

$$= (2.5 + 2.6) \div 0.75 + (0.75)$$

$$= 6.8 + 0.75 = 7.55 = \frac{755}{100} = \frac{151}{20}$$

$$\text{Sol.60.(a)} \left(\frac{5}{6} + \frac{3}{4}\right) \div \left(\frac{7}{8} - \frac{2}{3}\right)$$

$$= \left(\frac{10+9}{12}\right) \div \left(\frac{21-16}{24}\right)$$

$$= \frac{19}{12} \div \frac{5}{24} \Rightarrow \frac{19}{12} \times \frac{24}{5} = \frac{38}{5}$$

$$\text{Sol.61.(c)} \sqrt{45} + \sqrt{20} - \sqrt{80}$$

$$\sqrt{9 \times 5} + \sqrt{4 \times 5} - \sqrt{16 \times 5}$$

$$3\sqrt{5} + 2\sqrt{5} - 4\sqrt{5} = \sqrt{5}$$

$$\text{Sol.62.(b)} (0.04^3 + 0.01^3) \div (0.16^3 + 0.04^3)$$

$$(0.04^3 + 0.01^3) \div (0.16^3 + 0.04^3)$$

$$= \frac{(0.04^3 + 0.01^3)}{(0.16^3 + 0.04^3)} = \frac{(0.04^3 + 0.01^3)}{4^3 [0.04^3 + 0.01^3]}$$

$$= \frac{1}{4^3} = \frac{1}{64} \approx 0.015$$

$$\text{Sol.63.(a)} 4.625 = 4 + 0.625$$

$$\Rightarrow 0.625 = \frac{625}{1000} = \frac{5}{8}$$

$$\text{So, } 4.625 = 4 + \frac{5}{8} = \frac{37}{8}$$

$$\text{Sol.64.(b)} 7.4 - \left(2\frac{7}{8} + 2.75\right)$$

$$= 7.4 - (2.875 + 2.75)$$

$$= 7.4 - 5.625 = 1.775$$

$$\text{Sol.65.(b)} \frac{1}{16} + \frac{1}{32} = \frac{2}{32} + \frac{1}{32}$$

$$= \frac{3}{32} = 0.09375$$

$$\text{Sol.66.(b) Given :-} \sqrt{(14 + 6\sqrt{5})}$$

We can write,

$$= \sqrt{(\sqrt{9})^2 + (\sqrt{5})^2 + 2 \times \sqrt{9} \times \sqrt{5}}$$

$$= \sqrt{(\sqrt{9} + \sqrt{5})^2} = \sqrt{9} + \sqrt{5} = (3 + \sqrt{5})$$

$$\text{Sol.67.(a)} 42^3 + 45^3 - 63^3 + 149$$

$$74088 + 91125 - 250047 + 149$$

$$165362 - 250047 = -84685$$

Sol.68.(a) Given:-

$$(2.8 + 5.125) \times 0.45 - (0.90 \div 0.3)$$

$$(7.925) \times 0.45 - (3)$$

$$3.56625 - 3 = 0.566$$

Sol.69.(d)

$$\text{Given:-} (0.03^3 + 0.002^3) \div (0.3^3 + 0.02^3)$$

$$= \frac{(0.1)^3 (0.3^3 + 0.02^3)}{(0.3^3 + 0.02^3)} = \frac{1}{1000} = 0.001$$

Sol.70.(d)

$$\sqrt{a + \sqrt{a + \sqrt{a + \sqrt{a + \dots \infty}}}}$$

$$= \frac{\sqrt{4a+1}+1}{2}$$

$$\sqrt{30 + \sqrt{30 + \sqrt{30 + \sqrt{30 + \dots \infty}}}}$$

$$= \frac{\sqrt{4 \times 30 + 1} + 1}{2} = \frac{\sqrt{121} + 1}{2}$$

$$= \frac{11+1}{2} = 6$$

Short Trick:-

Co - prime factors of 30 = (5, 6)



In case of (+ve) sign, larger number is the right answer.

Sol.71.(a) According to the question,
 $18^3 + 20^3 - 29^3 + 116$
 $= 5832 + 8000 - 24389 + 116$
 $\Rightarrow (5832 + 8000 + 116) - 24389 = -10441$

Sol.72.(d) $(\sqrt{5} + 3)^2 = ? + 6\sqrt{5}$
 $\Rightarrow 5 + 9 + 2 \times \sqrt{5} \times 3 = ? + 6\sqrt{5}$
 $\Rightarrow 14 + 6\sqrt{5} = ? + 6\sqrt{5} \Rightarrow ? = 14$

Sol.73.(c) $(\sqrt{7,396} - \sqrt{1,156}) \times (\frac{1}{2})$
 $(86 - 34) \times (\frac{1}{2}) = 52 \times \frac{1}{2} = 26$

Sol.74.(a) Let, $a = 0.11$ and $b = 0.08$
 $\frac{(0.11^3 + 0.08^3)}{(0.22^3 + 0.16^3)} = \frac{(a^3 + b^3)}{2^3(a^3 + b^3)}$
 $= \frac{1}{8} = 0.125$

Sol.75.(c) $\frac{\{(0.6)^3 + (0.06)^3 + (0.006)^3\}}{\{(0.2)^3 + (0.02)^3 + (0.002)^3\}}$
Let, $0.006 = x$ and $0.002 = y$
Then
 $= \frac{(100x)^3 + (10x)^3 + x^3}{(100y)^3 + (10y)^3 + y^3}$
 $= \frac{x^3(100^3 + 10^3 + 1)}{y^3(100^3 + 10^3 + 1)}$
 $= \frac{x^3}{y^3} = (\frac{0.006}{0.002})^3 = 3^3 = 27$

Sol.76.(c) $\sqrt{711 + \sqrt{303 + \sqrt{441}}}$
 $\Rightarrow \sqrt{711 + \sqrt{303 + 21}}$
 $\Rightarrow \sqrt{711 + \sqrt{324}}$
 $\Rightarrow \sqrt{711 + 18} = \sqrt{729} = 27$

Sol.77.(d) $(0.09^3 + 0.03^3) \div (0.3^3 + 0.1^3)$
 $= \frac{0.3^3(0.3^3 + 0.1^3)}{(0.3^3 + 0.1^3)} = 0.027$

Sol.78.(c) $(\sqrt{6400} - \sqrt{2401}) \times (\frac{1}{2})$
 $= (80 - 49) \times (\frac{1}{2}) = 31 \times \frac{1}{2} = 15.5$

Sol.79.(d) $(\sqrt{5776} - \sqrt{4761}) \times (\frac{1}{3})$
 $= (76 - 69) \times (\frac{1}{3}) = 7 \times \frac{1}{3} = 2.3$

Sol.80.(b)
 $\frac{(0.16^3 + 0.04^3)}{(0.4^3 + 0.1^3)} = \frac{0.4^3[0.4^3 + 0.1^3]}{(0.4^3 + 0.1^3)}$
 $= 0.4^3 = 0.064$

Sol.81.(a)
 $(0.02^3 + 0.002^3) \div (0.04^3 + 0.004^3)$
 $= \frac{(0.02^3 + 0.002^3)}{(0.04^3 + 0.004^3)}$
Let $a = 0.02$ and $b = 0.002$
 $= \frac{a^3 + b^3}{(2a)^3 + (2b)^3} = \frac{a^3 + b^3}{8(a^3 + b^3)} = \frac{1}{8}$

Sol.82.(a) $\sqrt{10 + 4\sqrt{6}}$
On squaring

$10 + 4\sqrt{6} = 6 + 4 + 2\sqrt{6 \times 4} = (\sqrt{6} + \sqrt{4})^2$
On taking the square root
 $(\sqrt{6} + \sqrt{4}) = \sqrt{6} + 2$

Sol.83.(b) $17^3 + 20^3 - 19^3 + 346$
 $4913 + 8000 - 6859 + 346$
 $13259 - 6859 = 6400$

Sol.84.(a) If $a + b + c = 0$,
then, $a^3 + b^3 + c^3 = 3abc$
 $\frac{(0.4^3 + 0.3^3 - 0.7^3)}{(3 \times 0.4 \times 0.3 \times 0.7)} = \frac{-0.252}{0.252} = -1$

Sol.85.(a) $(\sqrt{13} + \sqrt{5})^2 - (\sqrt{13} - \sqrt{5})^2$
 $= 4\sqrt{13} \times \sqrt{5} = 4\sqrt{65}$

Sol.86.(b)
 $\frac{(3p)^3 + (3q)^3}{p^3 + q^3} = \frac{27(p)^3 + (q)^3}{p^3 + q^3} = 27$

Sol.87.(d) $\frac{1}{\sqrt{5} - \sqrt{2}} + \frac{1}{\sqrt{5} + \sqrt{2}}$
 $= \frac{\sqrt{5} + \sqrt{2} + \sqrt{5} - \sqrt{2}}{(\sqrt{5} - \sqrt{2})(\sqrt{5} + \sqrt{2})} = \frac{2\sqrt{5}}{3}$

Sol.88.(d) $(0.3^3 + 0.03^3) \div (0.6^3 + 0.06^3)$
 $= \frac{(0.3^3 + 0.03^3)}{(0.6^3 + 0.06^3)} = \frac{3^3(0.1^3 + 0.01^3)}{6^3(0.1^3 + 0.01^3)}$
 $= \frac{27}{216} = \frac{1}{8} = 0.125$

Sol.89.(a) $14.345 \times 0.037 \approx 0.5307$

Sol.90.(a) Given :- $\sqrt{9409} - \sqrt{1849}$
 $= 97 - 43 = 54$

Sol.91.(a) $\sqrt{20} = 2\sqrt{5}$, $\sqrt{80} = 4\sqrt{5}$
 $2\sqrt{5} + 4\sqrt{5} = 6\sqrt{5}$

Sol.92.(d) $8.65 + 9.28 + 0.067 = 17.997$

Sol.93.(c) $\sqrt{13 - 2\sqrt{42}} - \frac{1}{\sqrt{13 - 2\sqrt{42}}}$

$\sqrt{13 - 2\sqrt{42}}$
 $= \sqrt{(\sqrt{7})^2 + (\sqrt{6})^2 - 2\sqrt{7}\sqrt{6}}$
 $= \sqrt{(\sqrt{7} - \sqrt{6})^2} = \sqrt{7} - \sqrt{6}$
 $\frac{1}{\sqrt{13 - 2\sqrt{42}}} = \frac{1}{\sqrt{7} - \sqrt{6}}$
 $= \frac{\sqrt{7} + \sqrt{6}}{(\sqrt{7} - \sqrt{6})(\sqrt{7} + \sqrt{6})} = \sqrt{7} + \sqrt{6}$
Therefore,

$(\sqrt{7} - \sqrt{6}) - (\sqrt{7} + \sqrt{6}) = -2\sqrt{6}$

Sol.94.(a) $\frac{0.02505}{23^2 + 6^2 - 64}$
 $= \frac{0.02505}{529 + 36 - 64} = \frac{0.02505}{501} = 5 \times 10^{-5}$

Sol.95.(a) Given : $(\sqrt{8} - 1)^2 = ? - 2\sqrt{8}$

$\Rightarrow 8 + 1 - 2\sqrt{8} = ? - 2\sqrt{8}$
 $\Rightarrow 9 - 2\sqrt{8} = ? - 2\sqrt{8} \Rightarrow ? = 9$

Sol.96.(b) $\frac{\sqrt{7+4\sqrt{3}}}{\sqrt{7-4\sqrt{3}}}$
 $= \sqrt{\frac{7+4\sqrt{3}}{7-4\sqrt{3}}} \times \frac{7+4\sqrt{3}}{7+4\sqrt{3}} = 7 + 4\sqrt{3}$

Sol.97.(a) By option

$a = \frac{6}{16} = 0.375$, $b = \frac{2}{12} = 0.166666$
 $c = \frac{4}{14} = 0.285714$, $d = \frac{8}{18} = 0.444444$

Sol.98.(b)

$(\sqrt{7} - 2)^2 = 7 + 4 - 4\sqrt{7} = 11 - 4\sqrt{7}$
We compare equations,
then $\Rightarrow 11 - 4\sqrt{7} = ? - 4\sqrt{7} \Rightarrow ? = 11$

Sol.99.(a)

$(\sqrt{5} + \sqrt{3})^2 = (\sqrt{5})^2 + (\sqrt{3})^2 + 2\sqrt{5} \times \sqrt{3}$
 $= 5 + 3 + 2\sqrt{15} = 8 + 2\sqrt{15}$

Sol.100.(b)

$\frac{\sqrt{16+2\sqrt{55}} + \sqrt{16-2\sqrt{55}}}{\sqrt{4}} =$
 $\frac{\sqrt{(\sqrt{11})^2 + (\sqrt{5})^2 + 2 \times \sqrt{11} \times \sqrt{5}} + \sqrt{(\sqrt{11})^2 + (\sqrt{5})^2 - 2 \times \sqrt{11} \times \sqrt{5}}}{\sqrt{4}}$
 $= \frac{\sqrt{(\sqrt{11} + \sqrt{5})^2} + \sqrt{(\sqrt{11} - \sqrt{5})^2}}{\sqrt{4}}$
 $= \frac{\sqrt{11} + \sqrt{5} + \sqrt{11} - \sqrt{5}}{\sqrt{4}} = \sqrt{11}$

Sol.101.(a) $\sqrt{4 \times 225 \div 15 + 25 - 4}$
 $= \sqrt{4 \times 15 + 25 - 4}$
 $= \sqrt{60 + 25 - 4} = \sqrt{85 - 4} = \sqrt{81}$

Sol.102.(b) $\sqrt{(11 + 4\sqrt{6})}$
On squaring

$(11 + 4\sqrt{6}) = 8 + 3 + 2\sqrt{8 \times 3}$
 $= (\sqrt{8} + \sqrt{3})^2$

On taking square root
 $(\sqrt{8} + \sqrt{3}) = 2\sqrt{2} + \sqrt{3}$

Sol.103.(b) $\frac{?}{\sqrt{64}} = \frac{\sqrt{121}}{10 + 1}$

$\frac{?}{8} = \frac{11}{11} \Rightarrow ? = 1 \times 8 = 8$

Sol.104.(c) $(\sqrt{6241} - \sqrt{1260.25}) \times (\frac{1}{6})$
 $= (79 - 35.5) \times \frac{1}{6} = 43.5 \times \frac{1}{6} = 7.25$

Sol.105.(d) $2\sqrt{5}(\sqrt{45} - \sqrt{20})$
 $= 2\sqrt{5} \times \sqrt{5}(3 - 2) = 2 \times 5 = 10$

Sol.106.(b) $\frac{\sqrt{6} + \sqrt{7}}{\sqrt{6} - \sqrt{7}} \times \frac{\sqrt{6} + \sqrt{7}}{\sqrt{6} + \sqrt{7}}$
 $= \frac{(\sqrt{6} + \sqrt{7})^2}{(\sqrt{6})^2 - (\sqrt{7})^2} = \frac{6 + 7 + 2 \times \sqrt{6} \times \sqrt{7}}{-1}$



$$= -13 - 2\sqrt{42}$$

Sol.107.(b) Given ,

$$\sqrt{(8 \times 447 \div 24 + 75 - 28)} \\ = \sqrt{(149 + 75 - 28)} = \sqrt{196} = 14$$

Sol.108.(a) $(12.12)^2 - 396$

$$= (12 + 0.12)^2 - 396$$

$$= 12^2 + 2(12)(0.12) + (0.12)^2 - 396$$

$$= 144 + 2.88 + 0.0144 - 396 = -249.1056$$

Sol.109.(a)

$$9\frac{1}{5} - \left[\frac{5}{6} \div \left\{ \frac{5}{6} - \left(\frac{1}{2} \times \left(\frac{7}{4} - \frac{1}{4} \right) \right) \right\} \right]$$

$$\frac{46}{5} - \left[\frac{5}{6} \div \left\{ \frac{5}{6} - \left(\frac{1}{2} \times \left(\frac{6}{4} \right) \right) \right\} \right]$$

$$\frac{46}{5} - \left[\frac{5}{6} \div \left\{ \frac{5}{6} - \frac{3}{4} \right\} \right]$$

$$\frac{46}{5} - \left[\frac{5}{6} \div \left\{ \frac{1}{12} \right\} \right]$$

$$\frac{46}{5} - \left[\frac{5}{6} \times \frac{12}{1} \right] \Rightarrow \frac{46}{5} - 10 = -\frac{4}{5}$$

Sol.110.(a) $x = 5 + \frac{1}{5 + \frac{1}{6 + \frac{1}{2}}}$

$$\Rightarrow x = 5 + \frac{1}{5 + \frac{2}{13}}$$

$$\Rightarrow x = 5 + \frac{13}{67} \Rightarrow x = \frac{348}{67}$$

Sol.111.(a) Given :-

$$x = 3 + \frac{1}{8 + \frac{1}{6 + \frac{1}{2}}} \Rightarrow x = 3 + \frac{1}{8 + \frac{2}{13}}$$

$$x = 3 + \frac{13}{106} \Rightarrow x = \frac{331}{106}$$

Sol.112.(b) On checking all options one by one, only option (b) satisfies ,

$$= (\sqrt{5} + \sqrt{5})^2 = (2\sqrt{5})^2 = 4 \times 5 = 20$$

Sol.113.(b)

$$\frac{9\sqrt{3}}{3\sqrt{3}} = \frac{9}{3} \times \frac{\sqrt{3}}{\sqrt{3}} = 3 \times 1 = 3$$

Sol.114.(a) $\frac{253}{16} = 15.8125$

Sol.115.(b)

$$\sqrt{0.0049} = \sqrt{\frac{49}{10000}} = \frac{7}{100} = 0.07$$

Sol.116.(b)

Reciprocal of $\frac{5}{7} = \frac{1}{\frac{5}{7}} = \frac{7}{5}$

Sol.117.(a) $\frac{9}{20} = 0.45$

Sol.118.(b) $34 \times 34 = 1156$

Sol.119.(a) $23 \times 23 = 529$

Sol.120.(b) $0.6 < \frac{3}{4} \Rightarrow 0.6 < 0.75$

Sol.121.(a) Given :- $0.4757575... = 0.\overline{475}$

$$\frac{475 - 4}{990} = \frac{471}{990} = \frac{157}{330}$$

Sol.122.(d) Given :- $0.4 = \frac{4}{10} = \frac{2}{5}$

Sol.123.(c) $\sqrt{18} + \sqrt{50}$

$$\sqrt{9 \times 2} + \sqrt{25 \times 2}$$

$$3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$$

Sol.124.(b) $\sqrt{(7 + 4\sqrt{3})}$

$$= \sqrt{(\sqrt{4})^2 + (\sqrt{3})^2 + 2\sqrt{4} \cdot \sqrt{3}}$$

$$= \sqrt{(\sqrt{4} + \sqrt{3})^2} = \sqrt{4} + \sqrt{3} = 2 + \sqrt{3}$$

Sol.125.(a) $\sqrt{(9 + 4\sqrt{5})}$

$$= \sqrt{(\sqrt{4})^2 + (\sqrt{5})^2 + 2\sqrt{4} \cdot \sqrt{5}}$$

$$= \sqrt{(\sqrt{4} + \sqrt{5})^2} = (\sqrt{4} + \sqrt{5})$$

Sol.126.(b) Given, $(\sqrt{5} + \sqrt{7})^2$

$$\Rightarrow 5 + 7 + 2\sqrt{7 \times 5} \Rightarrow 12 + 2\sqrt{35}$$

Sol.127.(b) $(0.96 \div 8) \div (1.30 - 0.34)$

$$= \frac{0.12}{0.96} = \frac{1}{8}$$

Sol.128.(b) $\sqrt{8 + 2\sqrt{15}} + \sqrt{8 - 2\sqrt{15}}$

$$= \sqrt{(\sqrt{5} + \sqrt{3})^2} + \sqrt{(\sqrt{5} - \sqrt{3})^2}$$

$$= \sqrt{5} + \sqrt{3} + \sqrt{5} - \sqrt{3} = 2\sqrt{5}$$

Sol.129.(c) $\left(\frac{1}{21}\right) + \left(\frac{1}{42}\right) + \left(\frac{1}{63}\right) + \left(\frac{1}{28}\right)$

$$= \frac{12 + 6 + 4 + 9}{252} = \frac{31}{252} = 0.12301$$

Sol.130.(c)

$$\left(\frac{1}{21}\right) + \left(\frac{1}{28}\right) + \left(\frac{1}{35}\right) + \left(\frac{1}{42}\right)$$

$$\frac{1}{3 \times 7} + \frac{1}{4 \times 7} + \frac{1}{5 \times 7} + \frac{1}{6 \times 7}$$

$$\frac{1}{7} \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} \right)$$

$$\frac{1}{7} \left(\frac{20 + 15 + 12 + 10}{60} \right)$$

$$\frac{1}{7} \times \frac{57}{60} = 0.1357$$

Sol.131.(a)

$$\left(5\frac{1}{2} + 3.75\right) \times \left(\frac{3}{8} - 0.125\right) \div \left(1\frac{2}{5} + \frac{4}{10}\right)$$

$$\left(\frac{11}{2} + \frac{15}{4}\right) \times (0.375 - 0.125) \div \left(\frac{7}{5} + \frac{2}{5}\right)$$

$$\left(\frac{37}{4}\right) \times 0.25 \div \left(\frac{9}{5}\right)$$

$$\frac{37}{4} \times \frac{1}{4} \times \frac{5}{9} = \frac{185}{144}$$

Sol.132.(c) $\sqrt{108} + \sqrt{12} - \sqrt{75}$

$$= 6\sqrt{3} + 2\sqrt{3} - 5\sqrt{3} = 3\sqrt{3}$$

Sol.133.(b) $(4\sqrt{7} + 3\sqrt{28}) - (\sqrt{63} + 5\sqrt{7})$

$$(4\sqrt{7} + 6\sqrt{7}) - (3\sqrt{7} + 5\sqrt{7})$$

$$10\sqrt{7} - 8\sqrt{7} = 2\sqrt{7}$$

Sol.134.(d) $(\sqrt{2} + \sqrt{7})^2$

$$= 2 + 7 + 2\sqrt{2 \times 7} = 9 + 2\sqrt{14}$$

Sol.135.(a)

$$\left(\frac{1}{(7 - \sqrt{6})}\right) + \left(\frac{1}{(7 + \sqrt{6})}\right) - \left(\frac{14}{(49 - 6)}\right)$$

$$= \frac{7 + \sqrt{6} + 7 - \sqrt{6}}{7^2 - (\sqrt{6})^2} - \frac{14}{(49 - 6)}$$

$$= \frac{14}{(49 - 6)} - \frac{14}{(49 - 6)} = 0$$

Sol.136.(a) $0.2 + 0.22 + 0.222 = 0.642$

Sol.137.(c)

$$5 + 24 \times 8 \div 8 \text{ of } 8 - 8 \div 8 \times 4 \text{ of } 4$$

$$= 5 + 24 \times 8 \div 64 - 1 \times 16 = 5 + 3 - 16 = -8$$

Sol.138.(b)

$$(2z - 5y)^2 + (5z + 2y)^2 - 25z^2$$

$$= 4z^2 + 25y^2 - 20zy + 25z^2 + 4y^2 +$$

$$20zy - 25z^2 = 29y^2 + 4z^2$$

Sol.139.(b) According to question,

HCF of (221 and 488) = 1

Since the HCF is 1, the fraction $\frac{221}{488}$ is already in its lowest terms.

So, the lowest term is $\frac{221}{488}$

Sol.140.(b) $\frac{(x^2 - 9)}{x + 3} = \frac{(x^2 - 3^2)}{x + 3}$

$$= \frac{(x - 3)(x + 3)}{(x + 3)} = (x - 3)$$

Sol.141.(b)

$$\frac{(8.3)^3 + (9.2)^3 + (6.1)^3 - 3 \times 8.3 \times 9.2 \times 6.1}{(8.3)^2 + (9.2)^2 + (6.1)^2 - 8.3 \times 9.2 - 9.2 \times 6.1 - 6.1 \times 8.3}$$

Identity used:- $(a)^3 + (b)^3 + (c)^3 - 3abc = (a + b + c)[(a)^2 + (b)^2 + (c)^2 - ab - bc - ac]$

$$\frac{(8.3)^3 + (9.2)^3 + (6.1)^3 - 3 \times 8.3 \times 9.2 \times 6.1}{(8.3)^2 + (9.2)^2 + (6.1)^2 - 8.3 \times 9.2 - 9.2 \times 6.1 - 6.1 \times 8.3}$$

$$= 8.3 + 9.2 + 6.1 = 23.6$$

Sol.142.(a) $\frac{0.58 \times 0.58 \times 0.58 - 0.001}{0.58 \times 0.58 + 0.058 + 0.01}$

$$= \frac{(0.58)^3 - (0.1)^3}{[(0.58)^2 + 0.1 \times 0.58 + (0.1)^2]}$$

Using identity :

$$(a)^3 - (b)^3 = (a - b)(a^2 + b^2 + ab)$$

$$= \frac{(0.58 - 0.1)[(0.58)^2 + (0.1)^2 + 0.1 \times 0.58]}{[(0.58)^2 + 0.1 \times 0.58 + (0.1)^2]}$$

$$= 0.58 - 0.1 = 0.48$$

Sol.143.(d)

$$45 \div 3 + 4 \times A - 48 \div 24 + 4 = 1$$



$$15 + 4A - 2 + 4 = 1 \Rightarrow 17 + 4A = 1$$

$$4A = 1 - 17 \Rightarrow 4A = -16 \Rightarrow A = -4$$

Sol.144.(b) Formula used :

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$\frac{(4.2)^3 - (1.2)^3}{(4.2)^2 + 5.04 + (1.2)^2}$$

$$= \frac{(4.2 - 1.2)[(4.2)^2 + 5.04 + (1.2)^2]}{(4.2)^2 + 5.04 + (1.2)^2} = 3$$

Sol.145.(b)

Formula : $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$

Now, we have ;

$$(1.55^2 + 0.55^2 + 1.55 \times 0.55)$$

On multiplying and dividing the above expression by $(1.55 - 0.55)$, We get ;

$$\frac{(1.55 - 0.55)(1.552 + 0.552 + 1.55 \times 0.55)}{(1.55 - 0.55)}$$

$$= \frac{1.55^3 - 0.55^3}{1} = 1.55^3 - 0.55^3$$

Sol.146.(d)

$$[(2.5 + 1.5)^2 \div 4 \text{ of } \frac{1}{2} \times 8 - 1]$$

$$[(4)^2 \div 2 \times 8 - 1] = 8 \times 8 - 1$$

$$= 64 - 1 = 63$$

Sol.147.(a) $\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x-y)(y-z)(z-x)}$

We know that, when $a + b + c = 0$ then

$$(a)^3 + (b)^3 + (c)^3 = 3abc.$$

Now, $x - y + y - z + z - x = 0$

$$\text{So, } \frac{3(x-y)(y-z)(z-x)}{(x-y)(y-z)(z-x)} = 3$$

Sol.148.(d) $(g - h)^2 - (g + h)^2 + 4gh$

Identity used : $a^2 - b^2 = (a + b)(a - b)$

$$= (g - h - g - h)(g - h + g + h) + 4gh$$

$$= (-2h)(2g) + 4gh = -4gh + 4gh = 0$$

Sol.149.(d)

$$\frac{1.5 \times 1.5 + 2.5 \times 2.5 + 3.5 \times 3.5 + 2 \times 1.5 \times 2.5 + 2 \times 1.5 \times 3.5 + 2 \times 2.5 \times 3.5}{1.5 + 2.5 + 3.5}$$

We know that $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

$$\frac{(1.5 + 2.5 + 3.5)^2}{1.5 + 2.5 + 3.5} = 1.5 + 2.5 + 3.5 = 7.5$$

Sol.150.(a)

$$[(5\frac{1}{2} - 3\frac{1}{3}) \text{ of } 1\frac{5}{13} + 1 - 2 \times 3\frac{5}{2} \div 2\frac{3}{4}]$$

$$= [(\frac{13}{6}) \text{ of } \frac{18}{13} + 1 - 2 \times \frac{11}{2} \times \frac{4}{11}]$$

$$= [3 + 1 - 4] = 0$$

Sol.151.(b) $\frac{(4x^2 + 9y^2 - 12xy)}{(3y - 2x)}$

$$\frac{(2x)^2 + (3y)^2 - 2 \times 2x \times 3y}{(3y - 2x)}$$

$$\frac{(3y - 2x)^2}{(3y - 2x)} = 3y - 2x$$

Sol.152.(d)

$$25 \div 5 + 3 \text{ of } 2 - 4 + 6 \text{ of } 3 \div 2 - 8 + 4$$

$$= 5 + 6 - 4 + 18 \div 2 - 8 + 4$$

$$= 7 + 9 - 8 + 4 = 12$$

Sol.153.(b)

$$= 0.07 \times 0.28 \div 0.04 + 0.64 - 1.64 \div 0.04$$

$$= 0.07 \times 7 + 0.64 - 41$$

$$= 0.49 - 40.36 = -39.87$$

Sol.154.(c) $11 - \{(3 \div 4) \text{ of } 8 - 4 - 8\}$

$$\Rightarrow 11 - \{6 - (-4)\} \Rightarrow 11 - 10 = 11 - 10 = 1$$

Sol.155.(a) $133 \div 19 \times 3 - 48 \times 2 \div 24 + 841 \div [(45 - 16)]$

$$= 7 \times 3 - 48 \times \frac{2}{24} + 841 \div 29$$

$$= 21 - 4 + 29 = 46$$

Sol.156.(d)

$$9 \div [1 + \{4 \times (\frac{5}{6} - \frac{1}{3} + \frac{1}{2})\}]$$

$$9 \div [1 + \{4 \times (\frac{5}{6} - \frac{2+3}{6})\}]$$

$$9 \div [1 + \{4 \times (\frac{5}{6} - \frac{5}{6})\}]$$

$$9 \div [1 + \{4 \times 0\}] \Rightarrow 9 \div 1 = 9$$

Sol.157.(d) $9 \times N \text{ of } 4 \div 4 - 15 \times 4 = 12$

$$9 \times 4N \div 4 - 60 = 12$$

$$9N - 60 = 12 \Rightarrow 9N = 72 \Rightarrow N = 8$$

Sol.158.(a) $\frac{3(16^3 - 6^3)}{(16^2 + 6^2 + Q)} = 30$

$$\frac{3(4096 - 216)}{(256 + 36 + Q)} = 30$$

$$\frac{3(3880)}{(292 + Q)} = 30 \Rightarrow 388 = 292 + Q$$

$$Q = 388 - 292 = 96$$

Sol.159.(b) $(60 + 64 \div 4 \text{ of } 4) \times [1000 \div \{(10 \text{ of } (3 + 2)) \times 2 \text{ of } 5\} - 28 + (100 \times 3 \text{ of } 3)]$

$$= (60 + 64 \div 16) \times [1000 \div \{10 \text{ of } 5 \times 10\} - 28 + (100 \times 9)]$$

$$= (60 + 4) \times [1000 \div 500] - 28 + 900$$

$$= 64 \times 2 - 28 + 900$$

$$= 128 - 28 + 900 = 1000$$

Sol.160.(b) $626 \times 2912 \div 8138$

$$= 626 \times \frac{2912}{8138} = 626 \times \frac{112}{313}$$

$$= 2 \times 112 = 224$$

Sol.161.(d) $\frac{4x^2 - 49y^2}{2x + 7y} + 7y - 2x$

$$= \frac{(2x)^2 - (7y)^2}{2x + 7y} + 7y - 2x$$

$$\frac{(2x + 7y)(2x - 7y)}{2x + 7y} + 7y - 2x$$

$$= 2x - 7y + 7y - 2x = 0$$

Sol.162.(b) $\frac{7 + 8 \times 7 \div 8 + 7 \div 7 \div 7}{3 + 3 \times 3 \div 3 + 5 \div 5 \div 5}$

$$\frac{7 + 7 + 1 \div 7}{3 + 3 \times 1 + 1 \div 5} = \frac{14 + \frac{1}{7}}{6 + \frac{1}{5}}$$

$$= \frac{\frac{99}{7}}{\frac{31}{5}} = \frac{99}{7} \times \frac{5}{31} = \frac{495}{217} = 2\frac{61}{217}$$

Sol.163.(c) $\frac{4}{7} \times 4\frac{1}{2} \div 5\frac{1}{3} \text{ of } 2\frac{1}{2} - (7\frac{7}{8})$

$$\div 5\frac{1}{9} \text{ of } 9\frac{9}{20} + \frac{1}{2}$$

$$= \frac{4}{7} \times \frac{9}{2} \div \frac{16}{3} \text{ of } \frac{5}{2} - (\frac{63}{8} \div \frac{46}{9} \text{ of } \frac{189}{20}) + \frac{1}{2}$$

$$= \frac{4}{7} \times \frac{9}{2} \div \frac{80}{6} - (\frac{63}{8} \div \frac{483}{10}) + \frac{1}{2}$$

$$= \frac{4}{7} \times \frac{9}{2} \times \frac{6}{80} - (\frac{63}{8} \times \frac{10}{483}) + \frac{1}{2}$$

$$= \frac{27}{140} - (\frac{315}{1932}) + \frac{1}{2} = \frac{27}{140} - (\frac{105}{644}) + \frac{1}{2}$$

$$= \frac{97}{140} - (\frac{105}{644}) = \frac{97 \times 644 - 105 \times 140}{140 \times 644}$$

$$= \frac{62468 - 14700}{140 \times 644}$$

$$= \frac{47768}{140 \times 644} = \frac{4 \times 2 \times 7 \times 853}{140 \times 644} = \frac{853}{1610}$$

Sol.164.(c) $(3 - 3 \times 3 + 3 \div 3 + 3 \times 5) \times 2$

$$\text{of } 5 + (2 + 2 \div 2 + 2 \times 2 - 2)$$

$$= (3 - 9 + 1 + 15) \times 10 + (2 + 1 + 4 - 2)$$

$$= (19 - 9) \times 10 + (7 - 2)$$

$$= (10) \times 10 + (5) = 100 + 5 = 105$$

Sol.165.(a)

$$\{(3^3 + 2^3) \times \frac{22}{7}\} \times \{(6^2 + 3^2 - 4^2) \div (14\frac{1}{2})\}$$

$$= \{(27 + 8) \times \frac{22}{7}\} \times \{(36 + 9 - 16) \div (\frac{29}{2})\}$$

$$= \{35 \times \frac{22}{7}\} \times \{29 \div (\frac{29}{2})\}$$

$$= \{110\} \times \{29 \times \frac{2}{29}\} = \{110 \times 2\} \Rightarrow 220$$

Sol.166.(d) $[40 - 48 \div 4 \text{ of } 3] \div \{(6 + 4) - 4 \text{ of } 4 + 4 + (6 - 4) + (4 \times 4) \div 4\} \times 3$

$$= [40 - 48 \div 12] \div \{10 - 16 + 4 + 2 + 4\} \times 3$$

$$= 36 \div 4 \times 3 = 27$$

Sol.167.(d) $8 - 5 \div 2^3 \times 3 + (7 \div 7 \times 7) \div 7$

$$+ (3 \times 6 \div \frac{1}{3}) \times \frac{1}{3^3}$$

$$= 8 - 5 \div 8 \times 3 + (1 \times 7) \div 7 + (3 \times 6 \times 3)$$

$$\times \frac{1}{27}$$

$$= 8 - 5 \times \frac{1}{8} \times 3 + (7) \times \frac{1}{7} + (54) \times \frac{1}{27}$$

$$= 8 - \frac{15}{8} + 1 + 2 = 11 - \frac{15}{8} = \frac{73}{8}$$

Sol.168.(b)

$$10 + 3\frac{3}{5} \div [2\frac{2}{3} \div \{7\frac{1}{6} - (2\frac{1}{2} + \frac{2}{3})\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \div \{\frac{43}{6} - (\frac{5}{2} + \frac{2}{3})\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \div \{\frac{43}{6} - (\frac{15+4}{6})\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \div \{\frac{43-19}{6}\}]$$

$$= 10 + \frac{18}{5} \div [\frac{8}{3} \times \{\frac{6}{24}\}]$$

$$= 10 + \frac{18}{5} \div [\frac{2}{3}]$$

$$= 10 + \frac{18}{5} \times \frac{3}{2} = 10 + \frac{27}{5} = \frac{77}{5}$$

Sol.169.(b)



$$\begin{aligned}
 & 3\frac{4}{7} - [16\frac{1}{2} \div \{6^2 - (7 - 2\frac{1}{2})\}] \\
 &= \frac{25}{7} - [\frac{33}{2} \div \{36 - (7 - \frac{5}{2})\}] \\
 &= \frac{25}{7} - [\frac{33}{2} \div \{36 - (\frac{9}{2})\}] \\
 &= \frac{25}{7} - [\frac{33}{2} \div \frac{63}{2}] \\
 &= \frac{25}{7} - [\frac{33}{2} \times \frac{2}{63}] = \frac{25}{7} - [\frac{33}{63}] \\
 &= \frac{25 \times 9}{7 \times 9} - \frac{33}{63} \\
 &= \frac{225 - 33}{63} = \frac{192}{63} = \frac{64}{21}
 \end{aligned}$$

Sol.170.(a)

$$\begin{aligned}
 &= \frac{\frac{1}{2} \div \frac{3}{4} \text{ of } \frac{5}{6}}{\frac{1}{2} \div \frac{3}{4} \times \frac{5}{6}} + \frac{2 + 2 \times 2}{2 \div 2 \times 2} \\
 &= \frac{\frac{1}{2} \div \frac{15}{24}}{\frac{2}{3} \times \frac{5}{6}} + \frac{2 + 4}{2} = \frac{\frac{12}{15}}{\frac{5}{9}} + 3 \\
 &= \frac{12 \times 3}{5 \times 5} + 3 = \frac{36}{25} + 3 \\
 &= \frac{36 + 75}{25} = \frac{111}{25}
 \end{aligned}$$

Sol.171.(c) $4 \times 7 - 3\{7 \times 4 \div (6 \times 2)\}$
 $= 28 - 3\{7 \times 4 \div 12\}$
 $= 28 - 3\{\frac{7}{3}\} = 28 - 7 = 21$

Sol.172.(c) $420 \div [\frac{4}{9} \times (1 + \frac{4}{5}) \times 5] +$
 $[\frac{5}{7} \text{ of } \frac{21}{25} \div \frac{3}{4} \text{ of } \frac{2}{5}]$
 $= 420 \div [\frac{4}{9} \times (\frac{9}{5}) \times 5] + [\frac{3}{5} \div \frac{3}{10}]$
 $= 420 \div 4 + [\frac{3}{5} \times \frac{10}{3}] = 105 + 2 = 107$

Sol.173.(c) $\frac{2 \times 2 + 2}{2 \div 2 \times 2} \div \frac{\frac{1}{2} \div \frac{1}{2} \text{ of } \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \text{ of } \frac{1}{2}}$
 $\frac{4 + 2}{1 \times 2} \div \frac{\frac{1}{2} \div \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}} = 3 \div \frac{\frac{1}{2} \times 4}{\frac{6}{8}}$
 $3 \div \frac{2 \times 8}{6} = 3 \times \frac{6}{16} = \frac{9}{8} = 1\frac{1}{8}$

Sol.174.(d) $(24 \div 3 \text{ of } \frac{1}{6}) \times (\frac{4}{5} \div \frac{9}{8} \times$
 $\frac{3}{5}) \div (\frac{4}{7} \div \frac{3}{14} \text{ of } \frac{3}{14})$
 $= (24 \div \frac{1}{2}) \times (\frac{4}{5} \times \frac{8}{9} \times \frac{3}{5}) \div (\frac{4}{7} \div \frac{9}{196})$
 $= (24 \times 2) \times (\frac{32}{75}) \div (\frac{4}{7} \times \frac{196}{9})$
 $= 48 \times (\frac{32}{75}) \div (\frac{112}{9})$
 $= 48 \times (\frac{32}{75}) \times (\frac{9}{112})$
 $= 48 \times (\frac{2}{25}) \times (\frac{3}{7}) = \frac{288}{175}$

Sol.175.(b)

$$= \frac{5 + 7 \times 5 \div 8 \text{ of } 5 + 3 \div 3 \times 5}{6 \div 6 \text{ of } 4 + 9 \times 9 \div 3 \times 2 \times 2}$$

$$\begin{aligned}
 &= \frac{5 + 7 \times 5 \div 40 + 1 \times 5}{6 \div 24 + 27 \times 2 \times 2} \\
 &= \frac{\frac{7}{8} + 10}{\frac{1}{4} + 108} = \frac{\frac{7+80}{8}}{\frac{1+432}{4}} \\
 &= \frac{\frac{87}{8}}{\frac{433}{4}} = \frac{87 \times 4}{8 \times 433} = \frac{87}{866}
 \end{aligned}$$

Sol.176.(b) $11\frac{1}{4} \div 3\frac{3}{5} \times 6\frac{2}{3} + \frac{1}{6}$
 $= \frac{45}{4} \div \frac{18}{5} \times \frac{20}{3} + \frac{1}{6}$
 $= \frac{45}{4} \times \frac{5}{18} \times \frac{20}{3} + \frac{1}{6}$
 $= \frac{125}{6} + \frac{1}{6} = \frac{126}{6} = 21$

Sol.177.(c)

$$\begin{aligned}
 & [3^3 \{(40 \times 5) + (7 \times 6)\} \div 11] + (2^3 \times 15 \div 24) \\
 &= [27 \{(200 + 42) \div 11\} + 5] \\
 &= [27 \{242 \div 11\} + 5] \\
 &= [27 \times 22] + 5 = 594 + 5 = 599
 \end{aligned}$$

Sol.178.(d) Given expression :-

$$\begin{aligned}
 & 47 \div (10 \times 5) + (4 - 5) - 6 \\
 & \text{Replacing the signs as per the directions} \\
 & \text{given, we get;} \\
 & 47 \times (10 \div 5) - (4 + 5) + 6 \\
 &= 47 \times 2 - 9 + 6 = 94 - 3 = 91
 \end{aligned}$$

Sol.179.(a)

$$\begin{aligned}
 & \frac{[\frac{1}{4} \div \frac{1}{4} \times \frac{1}{4}] \times \frac{1}{4}}{[\frac{1}{8} + \frac{1}{8} \times \frac{1}{8}]} + (1 - \frac{4}{9}) \\
 &= \frac{\frac{1}{4} \times \frac{1}{4}}{\frac{1}{8} + \frac{1}{64}} + \frac{5}{9} = \frac{\frac{1}{16}}{\frac{9}{64}} + \frac{5}{9} \\
 &= \frac{1}{16} \times \frac{64}{9} + \frac{5}{9} = \frac{4}{9} + \frac{5}{9} = \frac{9}{9} = 1
 \end{aligned}$$

Sol.180.(d) $3\frac{1}{5} \text{ of } 2\frac{3}{4} \div \frac{11}{3} + (2\frac{1}{5} \div 1$

$$\begin{aligned}
 & \frac{2}{5} \text{ of } 1\frac{3}{7} - 2\frac{1}{3}) \\
 &= \frac{16}{5} \text{ of } \frac{11}{4} \div \frac{11}{3} + (\frac{11}{5} \div \frac{7}{5} \text{ of } \frac{10}{7} - \\
 & \frac{7}{3}) = \frac{12}{5} + (\frac{11}{10} - \frac{7}{3}) \\
 &= \frac{12}{5} - \frac{37}{30} = \frac{35}{30} = \frac{7}{6} = 1\frac{1}{6}
 \end{aligned}$$

Sol.181.(b)

$$\begin{aligned}
 & [18 - 3 \times \{3 - 3 \times (3 \div \frac{1}{3} - 3) + 3 \\
 & \div \frac{1}{3} + 3\} - 4 \times 2] \\
 &= [18 - 3 \times \{3 - 3 \times 6 + 12\} - 8] \\
 &= [18 - 3 \times (-3) - 8] = 18 + 9 - 8 = 19
 \end{aligned}$$

Sol.182.(c) $12 + 24 \div 2 - 7 \times 30 \div 6 + (5$
 $+ 4) \times 8 + 8 \div 8 - 8$
 $= 12 + 12 - 7 \times 5 + (9) \times 8 + 1 - 8$
 $= 12 + 12 - 35 + 72 + 1 - 8 = 97 - 43 = 54$

Sol.183.(a) $\frac{(143 \div 13 \times x - 3 \times 3)}{(6^2 - 7 \times 5 + x^2)} = 1$

$$\begin{aligned}
 & \Rightarrow \frac{11x - 9}{36 - 35 + x^2} = 1 \Rightarrow 11x - 9 = 1 + x^2 \\
 & \Rightarrow x^2 - 11x + 10 = 0 \Rightarrow x^2 - 10x - x + 10 = 0 \\
 & \Rightarrow x(x - 10) - 1(x - 10) = 0 \\
 & \Rightarrow (x - 10)(x - 1) = 0 \Rightarrow x = 10 \text{ or } 1 \\
 & \Rightarrow x = 1 \text{ (from option)}
 \end{aligned}$$

Sol.184.(b)

$$\begin{aligned}
 & \frac{12 + 3\frac{2}{3} \div [1\frac{3}{4} \div \{5\frac{1}{4} \times (3\frac{3}{5} \div 4\frac{1}{2})\}]}{1\frac{1}{3} \times 5\frac{1}{5}} \\
 &= \frac{12 + \frac{11}{3} \div [\frac{7}{4} \div \{\frac{21}{4} \times (\frac{18}{5} \div \frac{9}{2})\}]}{\frac{4}{3} \times \frac{26}{5}} \\
 &= \frac{12 + \frac{11}{3} \div [\frac{7}{4} \div \frac{21}{5}]}{\frac{104}{15}} \\
 &= \frac{12 + \frac{11}{3} \div \frac{5}{12}}{\frac{104}{15}} = \frac{12 + \frac{44}{5}}{\frac{104}{15}} \\
 &= \frac{\frac{104}{5}}{\frac{104}{15}} = \frac{104}{5} \times \frac{15}{104} = 3
 \end{aligned}$$

Sol.185.(c) Formula used :

$$\begin{aligned}
 & (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \\
 & 25 \times 25 + 9 \times 9 + 6 \times 6 + 2 \times 25 \times 9 + 2 \times 9 \\
 & \times 6 + 2 \times 25 \times 6 \\
 &= 25^2 + 9^2 + 6^2 + 2 \times 25 \times 9 + 2 \times 9 \times 6 + 2 \\
 & \times 25 \times 6 = (25 + 9 + 6)^2 = 1600
 \end{aligned}$$

